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**EXPLORING THE RELATIONSHIP BETWEEN INDEPENDENT
EXPENDITURES AND ELECTORAL OUTCOME**
**AN APPLICATION TO THE 2020 U.S. ELECTIONS FOR THE HOUSE
OF REPRESENTATIVES**

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Republicans defied the odds and grew our party last night.

Kevin McCarthy, House Republican Leader

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Introduction

Republicans, Independents, and Democrats alike are alarmed by money's growing role. Many are strongly opposed to "dark money" in elections [...] and disgruntled by the Citizens United ruling. - Craig Holman¹

The electoral process in the United States would seem to be very simple: every four years the President is re-elected, every two-thirds of the Senate and the entire House of Representatives, plus a series of electoral events every year, but in reality, this hides an even more crucial factor. The constant presence of elections, whether primary or general, induces legislators to behave as if they were always subjected to a new electoral challenge, forcing them, at times, to focus more on flagship programs and initiatives rather than legislation with a greater influence in the long term. This constant need to behave as if they were always in the election campaign involves, among other things, an ever-increasing need for electoral funds, both within the electoral campaigns and available to external committees that have the opportunity to support them, relaunching the centrality and the need for funds available to candidates for all U.S. elective offices. A clear example of the role played by campaign funds comes from a dispute over their counting between the two current leaders of the House of Representatives and well reported by Punchbowl News. On July 17, 2023, Punchbowl News, one of the more influential online political news dailies in the District of Columbia, stated that House Democratic Leader, Hakeem Jeffries, "*raised a huge \$29 million during the second quarter; [...] Jeffries' tally is \$7.3 million more than what Speaker Kevin McCarthy raised in Q2²*". But just a few days later, on July 25 the online political news had to go back and reverse this fact because the huge difference between the leaders of the two parties was caused by the fact that Jeffries added to his total also the amount raised by the DCCC, while the Republican Speaker of the House of Representatives does not make the same addition, meaning that he has raised more money than his adversary³. By reading these few lines it is possible to understand the central role that electoral funds play during electoral campaigns, so much so that they are considered one of the main supports for the victory of candidates both during primary and general elections. But the role of funds is not only crucial for

¹ (Holman 2016, <https://time.com/4182502/campaign-finance-reform/>)

² (Sherman and Cohen 2023, <https://punchbowl.news/archive/71723-punchbowl-news-am/>)

³ (Desiderio and Sherman 2023, <https://punchbowl.news/archive/72523-punchbowl-news-am/>)

electoral campaigns, as there are also several organizations outside the campaigns created with the sole purpose of collecting money from taxpayers, especially among the richest ones, with the aim of helping a candidate, or an ideological group, to get elected. In fact, especially after the famous sentence provided by the Supreme Court in *Citizen United versus FEC*⁴, the influence of money on the outcome of the elections has dramatically enlarged. When the Roberts Court published the sentence in that case, they decided the invalidity of the McCain-Feingold Act or Bipartisan Campaign Reform Act (BRCA)⁵. In fact, since 2010 the US political landscape has drastically changed, not only because of the natural exchange in the leadership of the parties, both in the White House and in Congress, but also because of the Supreme Court landmark decision regarding campaign finance laws and free speech: *Citizen United versus Federal Election Commission*. Thanks to this sentence, the majority of the Supreme Court has allowed the creation of outside entities that are allowed to get donations and spend without limitation during elections. And this thesis will focus on this element, starting from the revolution provided by the emerging presence of independent organizations, the purpose of this thesis is to observe their role in the electoral results for the House of Representatives in 2020. In order to better define the subject of the thesis, an introduction of the role that independent expenditures have gained in the last few years and may continue to have for a long time is much needed. In the '90s, two prominent members of the US Senate, John McCain (R-AZ) and Russ Feingold (D-WI) had called for a bill to reform the Federal Election Campaign Act of 1971⁶ and had tried to push for this type of legislation through the chambers of the Congress. In their first attempt, they failed to make the new bill pass the Senate floor, but after the 2000 election and the Enron scandal that shook up the presidential campaign of George W. Bush (Clarke 2005), an increasing number of people started to ask for reform as was proposed by the two senators some years before. Therefore, in 2002 politicians in both branches of Congress decided to act in order to reform the federal election legislation. There were many reasons to push for a reform of that legislation: not only the increasing problems that occurred during the previous presidential election but also the increasing power of external actors and lobbies that tried to influence the electoral

⁴ (*Citizens United versus Federal Election Commission* 2010)

⁵ (McCain and Feingold 2002)

⁶ (Mansfield 1972)

outcome by giving money to certain politicians instead of others, in order to help them in being elected. Moreover, another problem was emerging: the rapid increase in funds available to parties and electoral committees. Observing the data provided by the Federal Election Commission since 1982 about House candidates, it is possible to notice that between 1982 and 2002 there was an increase of more than 300% in the amount of money raised by people who ran both successfully and unsuccessfully for the lower chamber of Congress. The new law, named McCain-Feingold Act as the two main sponsors of this legislation, became effective on 6 November 2002 and was designed to address two issues: the growing influence of soft money in national campaigns and the proliferation of issue advocacy ads. The legal provisions of this new law have the purpose of modifying the precedent act passed by Congress following the Watergate scandal in 1974 and named Federal Election Campaign Act Amendments. The new law was the product of the work of both parties in both chambers of Congress and once President Bush received the bill, he signed it into law in March 2002. Shortly after this moment, the Bipartisan Campaign Reform Act started to face a lot of criticism, not only from free speech advocacy groups but also by some of the most important members of Congress like the Republican Whip in the Senate, Mitch McConnell. They argue that the new law would have limited free speech by interfering with the people's right to spend money in the electoral process while trying to advance certain politicians or policies. In fact, the bill faced at least four United States Supreme Court cases⁷ since 2003, and the first one was promoted right by McConnell because he argued that the law was unconstitutional because it limited the power of the First Amendment rights. The majority of the cases, once they came to the Supreme Court were dismissed, mainly because the Court based its decision on the opinion of the McConnell versus FEC case, if the holdings of the case were considered to be very confusing and the sentence is, also today, the longest in history. Therefore, the five elections after the bill became law were ruled through the use of the BCRA and the expenditures were subjected to the legal limits provided by this legislation. In 2008, the fifth and most notable case reached the United States Supreme Court,

⁷ (Davis versus Federal Election Commission 2008; Federal Election Commission versus Wisconsin Right to Life, Inc. 2007; McConnell versus Federal Election Commission 2003; Wisconsin Right to Life, Inc. versus Federal Election Commission 2006)

propelled by the conservative non-profit organization Citizens United⁸. The organization lodged a complaint in the U.S. District Court for the District of Columbia, contesting the constitutionality of various statutory provisions that regulate electioneering communications. On January 21, 2010, after the oral argument of both the parties of the trial, a majority of five justices, led by Justice Anthony Kennedy, sided with Citizens United in a decision that invalidated the restrictions on independent expenditures from corporate treasuries imposed by the BCRA. The Court held that these restrictions were in violation of the First Amendment's protection of free speech. The Court's decision had a far-reaching impact. It held that limiting corporate spending on independent political broadcasts during elections was an infringement on free speech rights. This ruling effectively opened the floodgates for increased corporate influence in political campaigns. One consequence of Citizens United was the creation of independent expenditures and their ability to accept unlimited contributions from every type of entity, both person and organization. These organizations, operating independently of candidates, could spend vast amounts of money to support or oppose political candidates, often inundating the airwaves with their advertisements. As the law, also the sentence faces a lot of criticism because it allowed the distortion of the democratic process by giving wealthy entities and individuals disproportionate influence over elections by enabling them to give unlimited money to a party or a Super PAC. Moreover, the critics expressed concerns about the potential for corruption, and the minority of the court, the four more liberal justices, wrote in the dissenting opinion that the majority decision was “*a rejection of the common sense of the American people, who have recognized a need to prevent corporations from undermining self-government*”⁹. On the other hand, the supporters of the decision like now Senate Minority Leader McConnell, saw it as a victory for free speech rights, arguing that corporations and organizations should have the same rights as individuals to express their political views through financial contributions, without any limitation provided by the interference of a law promoted by the US Congress. Since Citizens United, the case has remained a subject of ongoing debate and controversy. Efforts to mitigate its impact, such as proposing campaign finance reforms and pushing for a

⁸ (Citizens United versus Federal Election Commission 2010)

⁹ (Citizens United versus Federal Election Commission 2010)

constitutional amendment to overturn the decision, have emerged from those seeking to address what they perceive as its negative consequences on the democratic process, mainly because some studies have shown that the sentence has provided an increasing Republican advantage around four percentage points in total and ten or more percentage points in multiple states, especially those more competitive (Klupp, Mialon, and Williams 2016). Based on these premises, the analysis that this thesis aims to carry out is aimed at observing the role of the expenses made by all independent organizations during the November 2020 elections. The 2020 election marked a pivotal moment in American political history, shaping the landscape of the White House, Senate, and House of Representatives. To understand this result, it is necessary to observe what happened during the presidential elections of 2016 and the mid-term elections of 2018. The 2016 elections yielded a seismic political shift. In the United States, Donald Trump, a business magnate with no prior political experience, clinched an unexpected victory over seasoned politician Hillary Clinton. Trump's victory was attributed to his appeal to disenfranchised voters and promises of economic revitalization. Meanwhile, Brexit shocked Europe as the UK voted to exit the European Union, reflecting widespread anti-establishment sentiments. These outcomes underscored the power of populism and dissatisfaction with the status quo. The 2016 elections marked a turning point, redefining political norms and shaping global trajectories in the years to come¹⁰. Two years later, halfway through Trump's term, midterm elections took place, which in turn brought about some notable changes in the political system. The 2018 elections reshaped the political landscape¹¹. Democrats secured a majority in the House of Representatives, ending Republican control. This shift introduced a new dynamic, leading to policy gridlock and increased oversight of the Trump administration. In the Senate, Republicans expanded their majority, impacting judicial appointments. The elections highlighted deep partisan divides and a surge of diverse candidates, including a record number of women and minorities elected. Voter turnout reached historic levels, reflecting heightened civic engagement. The 2018 elections set the stage for continued debates on healthcare, immigration, and climate change, shaping subsequent political discourse. Both of these results have led

¹⁰ (CNN 2016, <https://edition.cnn.com/election/2016/results>)

¹¹ (CNN 2018, <https://edition.cnn.com/election/2018/results>)

to important changes to the political system both from the point of view of the actors in the field and from that of the increasing polarization of the system. In 2020, there was a new jolt to the politics of the United States. In a closely watched and highly contested race, Democratic nominee Joseph R. Biden Jr. emerged victorious, defeating the incumbent President Donald J. Trump¹². With a record-breaking voter turnout, the elections showed a country politically divided in half between the two main parties. In the Senate, Democrats managed to regain control by winning two runoff elections in Georgia, resulting in a 50-50 split with Vice President Harris holding the tie-breaking vote¹³. This slim majority enabled the Biden administration to advance its legislative agenda, confirming key appointments and passing significant bills, including pandemic relief and infrastructure investment. The House of Representatives saw Democrats maintain their majority, albeit with a reduced margin. Speaker Nancy Pelosi continued to wield considerable influence, steering legislative efforts to align with the administration's priorities. However, the slimmer majority prompted increased calls for bipartisanship and collaboration to pass meaningful legislation¹⁴. Therefore the 2020 elections gave surprising results because, at the same time, Republicans yielded control of the White House, because Joe Biden defeated Donald Trump in the presidential election, and of the Senate to the Democrats, but have gained some unexpected seats in the House of Representatives. Despite the fact that the margin of voting rewarded Democrats who obtained more than three percent more, the Republicans have gained fourteen seats in the lower chamber of Congress and given the other party a majority of just five seats, the smallest majority for the Democrats since 1942. This type of outcome was largely unforeseen by the strategists of both parties, but also by the main independent analysts, especially because no one expected the GOP Senate Conference to lose control of the high chamber, while even less expected the same party to gain some seats in the House, while it was losing the other branch of Congress and the White House at the same time¹⁵. A lot of elements could have influenced this type of outcome: the increasing number of women and minority candidates proposed to the electorate by the GOP, the presence of Donald Trump on

¹² (Siders, Kumar, and Cadelago 2020)

¹³ (Arkin, Desiderio, and Forgey 2021)

¹⁴ (Wasserman 2020)

¹⁵ (Galston 2020)

the ballot that could have emphasized high Republican turnout, and many others. But, thanks to a remarkable track record of historical spending in previous elections, it becomes increasingly plausible that such expenditures could have exerted a substantial influence on the overall electoral outcome. This fact can be noticed simply by looking at the frequency with which the name of the main independent organization, the Super PACs, is used by some of the major political news media (CNN, Fox News, The Hill, The New York Times, and The Washington Post), it is possible to notice how this appears more than ten thousand times in their articles, summing the results obtained in the search section of the newspapers mentioned in the previous lines. Notably, the expenditures made by external committees have reached unparalleled heights, leading to the hypothesis that they have assumed a prominent and pivotal role as one of the most influential elements shaping election results. Hence, the primary objective of this study will be to investigate whether the outcomes of the 2020 House of Representatives elections were impacted by the presence of outside expenditures within the political landscape of the United States. It is safe to say that not only did the expenditures made by independent organizations play a role in determining the outcome of those elections, but there were certainly many factors. For this reason, the work that will be carried out in the following chapters will use the expenses made by these organizations in favor of both Democrats and Republicans as the main independent variables but will take into account a whole series of other factors that could have been decisive results. In fact not only the independent expenditures seem to play a central role in determining the electoral outcome, and in addition to the expenses made by independent organizations, it is necessary to evaluate the role that the expenses made by the candidates' electoral campaigns, the partisan vote index referring to the districts, the result of the presidential elections in the districts, and the result of the previous electoral cycle will be used. The use of all these factors will be more fully justified within the first chapter of this work, but it can be argued that, given the literature used, each of the elements mentioned above could play a crucial role in the selected dependent variable, namely the differential between the votes obtained by the Republican Party and those obtained by the Democratic Party in each district during the elections for the House of Representatives in 2020. Moreover, all these variables will be explained in detail within the second chapter of this thesis, when,

through the use of descriptive statistics and graphs, each of the elements used will be explained together with the models used by the analysis, namely OLS regression, spatial regression, and IV regression. For these reasons, therefore, this work will be divided into three chapters, each of which will add a fundamental element to the success of the analysis that this thesis aims to do.

This thesis, therefore, will consist of the aforementioned three chapters, one for the revision of the previous literature on this topic, one for the presentation of the data and methodology used to build the models used, and one in which the models used will be exposed, also taking into account the limits and possible future developments of this work. The first chapter of this thesis will introduce a large part of the literature from which this thesis will start, by analyzing and evaluating a lot of topics in order to better define the subject of the research and the methods employed by previous political scientists to address the role of money in the United States election. The first element that will be considered will concern the structure of the electoral process, dividing the previous analyses between those who observed only the primary elections (Christenson and Smidt 2014), those who took into account both primaries and general elections (Lazarus 2005), and those who considered only the latter (T. Ferguson, Jorgensen, and Chen 2019), as done by this thesis. The difference between these processes is crucial in the analysis that the thesis wants to carry out as the primaries have a crucial role in the general elections. Following previous work on this subject and considering some substitute factors for the role of primaries, it is possible to analyze the role played by independent organizations only during general elections. As will be shown in the first chapter, there are three main factors that allow this operation to be carried out. Namely observing the result of the previous election cycle, taking into account the incumbent party instead of the member as an individual, and considering the competitiveness of the district using the Partisan Voting Index (PVI). The first factor could play an important role given the high re-election rate of incumbents, the second factor means that all districts can be considered equally, avoiding classifying districts with the incumbent representative retiring as open, and the third factor allows to evaluate the competitiveness of the electoral district, a crucial element in determining the outcome of the elections. After taking into account all these elements derived from the historical-political environment of the United States, the

analysis of the literature will shed light on a crucial difference regarding electoral expenses, namely that between those made by independent organizations and those made by the committees of candidates between which, and it will be fundamental for the thesis, there can be no kind of cooperation. Using the research done by Jacobson (1978, 2006, 2015), this distinction will be clarified, especially since the expenses made by the two main candidates in each district will be used as control variables within the models that will be developed. Referring to the immense amount of literature present in this field, this work will take into consideration what was written by Green and Krasno (1988), also by Gerber (1998), about the possible presence of a problem of endogeneity linked to the use of the electoral result as a dependent variable and the expenses made as an independent variable. The increase or decrease in expectations of victory, in fact, could lead to significant variations in the candidates' ability to raise funds and therefore their availability of spending. Reverse causality, as highlighted, is therefore based on expectations of the election result and could create some problems in the application of a regression model. For this reason, the decision taken was to also apply a model with instrumental variables (IV), using the expenses made by independent organizations during the previous election cycle, namely that of 2018, as instrument for the expenditures made during the analyzed electoral cycle, thanks to the fact that previous campaign spending does not directly affect the outcome of the following elections. By including the expenses made in support of both parties by independent organizations within the IV model it will be possible to overcome the problem of the presence of reverse causality that could have been highlighted in a model with an ordinary least square regression (OLS).

The analysis of the literature will then move to an exhaustive review of the role of independent expenditures and their classification according to the Federal Electoral Commission (FEC). Their role, as mentioned in the first pages of this thesis, has become increasingly central within the U.S. political system after the Supreme Court ruling in the *Citizen United versus FEC* case, radically changing the political landscape. The classification that the commission offers of these expenses is fundamental in the construction of the two main independent variables of this work as it will be the sum of the expenses made in favor of a candidate and in opposition of the opponent, divided according to the party to which the candidates belong. Given the

magnitude of the values obtained in these two variables, the expenses will then be deducted in millions of dollars, as shown in the second chapter, as a variation of a few tens would not have the possibility of moving a large number of votes while some millions do. Finally, the chapter will close with the presentation of the last elements that may have played a crucial role in determining the outcome of the elections, namely the presence of other simultaneous elections, the nationalization of electoral campaigns, and the sharing of television markets between different districts. The first of the three elements refers mainly to the presence of elections for state offices, such as those for the governor or one of the two senators, or the election of the President of the United States. In particular, the latter will be considered as a further element of control in the analysis because these could attract votes for a single party avoiding the presence of the phenomenon of ticket splitting (Burden and Kimball 1998). The second of the three elements refers to the process by which candidates tend to consider the challenge as if it had a national character, using messages referring to the leaders of the opposite camp, losing some of the contacts that the founding fathers had imagined for the House of Representatives (Carson and Hitefield 2018). The use of this type of electoral message allows a wider diffusion of the same and, at the same time, could help candidates of the same party in the districts nearby. This factor, combined with the sharing of television markets between neighboring districts, could help the electoral victory of one or the other party as a candidate would receive the support of the advertisements also of those of the neighboring districts. therefore, it would seem possible to hypothesize that a spatial regression model can be applied to these factors that will be more fully illustrated within the following chapters together with those previously mentioned, namely the OLS regression and the IV regression.

Following the literature review, the second chapter will be divided into two different sections, in the first one there will be the analysis of the data employed for the research, while in the second one, it will be possible to observe the structure of the models employed to achieve the purpose of the research. The first section of the chapter will be divided into eight paragraphs, equal to the number of variables used for the analysis conducted by this thesis. Each element will be thoroughly analyzed and exposed through the use of graphs and tables that will allow to better understand both how these are structured and their use within the models. The first two paragraphs

will concern expenditure, respectively that of independent organizations and those of candidates' committees. the third paragraph will analyze the PVI structured on the basis of the election results of the two previous presidential election cycles. The fourth paragraph will introduce the use of a possible spatial variable, as anticipated in the first chapter. The next three paragraphs will cover the election results of the House of Representatives in both 2020 and 2018, as well as those of the presidential elections in 2020. It is important to note that the election results will be considered by applying the two-party vote share formula, which is a differential between the votes obtained by Republicans and the votes obtained by Democrats. The last paragraph will expose the values of the instrumental variable that will be used, which are the expenses made by independent organizations in 2018.

After exposing the data used, in the second part of the chapter I will introduce the models employed in the analysis conducted by this thesis and also in this case each model will contain a different paragraph. The first paragraph will introduce the OLS regression that will use all the variables previously described. The second one will introduce the model with spatial regression by exposing the results of the Moran test which, since they are not sufficiently significant, will not allow to develop beyond this type of regression. The third paragraph will expose the model containing the regression with the instrumental variables that it will not consider the election result of the 2018 elections, contained only in the OLS regression, but that it will use the funds spent during that election cycle as a tool to assess the role of spending by independent organizations in 2020. Both the paragraph concerning the OLS regression and the one concerning the IV regression will also expose the statistical values that will be used by the analysis to evaluate the validity of the models used by this thesis. Due to the fact that the p-value of the Moran test conducted in this chapter turns out to be very low, it will not be necessary to apply a spatial regression as a model of this thesis and, therefore, the models that the third chapter will expose will be only the one with the OLS regression and the one with the IV. The third chapter, in addition to containing the results of both regressions, will try to interpret the results based on both what happened in reality and on the previous literature analyzed. The last paragraph of the chapter will contain the interpretation of the values and the justification of what has been achieved in the research conducted so far. Although the results show that the

expenditures made by independent organizations may have been insignificant in determining the composition of the House of Representatives, the analysis shows that all other variables, especially the outcome of the presidential election and the partisan leaning of each district, are highly significant and appear to have played a crucial role in determining the outcome of the 2020 election. Within this same paragraph will also be inserted the limitations of the research conducted so far as well as the possible future improvements of this research. Finally, a focus will be made on the use of spatial regression in this area, especially considering the fact that, thanks to the literature used, it could be useful to carry out research in this sense.

Chapter 1 – Literature Review

Any proponent of free speech should applaud this decision. Citizens United is and will be a First Amendment triumph of enduring significance. – Mitch McConnell¹⁶

Since money and politics have always had an incredibly special relationship, many studies have almost tried to approach this topic by trying to evaluate if and how much the amount of dollars spent influences the outcome of the elections. Especially after the United States Supreme Court judgment in the case Citizen United versus Federal Election Commission (FEC) in 2010, the amount of money spent by organizations and wealthy individuals, in an attempt to influence the outcome of the congressional elections, has dramatically increased reaching a new all-time high every election. For this reason, beginning in 2010, numerous scholars have earnestly sought to decipher the influence exerted by this emerging breed of independent expenditures and the funds they amass and employ (Hasson 2016; La Raja 2013; Ferguson 2015). Their focus extends beyond the presidential elections to encompass all other races, be it at the federal or local level. The reason for this comprehensive analysis stems from the understanding that each type of election bestows a certain degree of power in shaping the landscape of the United States public life. Consequently, academics have been driven to explore the extent to which these new independent entities, along with their financial resources, have impacted the outcomes of various elections. This broader perspective allows for a more comprehensive understanding of the overall influence of outside expenditures in shaping the political trajectory of the United States, extending beyond the realm of presidential contests. By examining federal and local races, researchers can delve into the nuanced dynamics and multifaceted nature of independent organizations' impact on the broader spectrum of United States elections. Most importantly, those previous studies should prove highly beneficial for the purpose of this thesis as they provide a solid groundwork from which to embark on extensive research. Especially after 2010, political analysts have started to observe the relationship between expenditures and the probability that a certain candidate has to win the general election. In fact, as La Raja (2014) notes, although the use of

¹⁶ (Mitch McConnell 2010, https://www.mcconnell.senate.gov/public/index.cfm?p=PressReleases&ContentRecord_id=78607ccf-6e92-4cfb-830a-9b25d89f4632&ContentType_id=c19bc7a5-2bb9-4a73-b2ab-3c1b5191a72b&Group_id=0fd6ddca-6a05-4b26-8710-a0b7b59a8f1f)

spending by independent organizations was innovative during the 2012 election cycle, by 2014 it was no longer seen as a novelty, but as an integral part of the U.S. political system. Many works focus on this topic using different approaches and methods to consider if the outcome of the elections was influenced by the expenditures made not only by each candidate but also by outside organizations, namely the Super PACs and other entities that made independent expenditures. Super PACs, or "Super Political Action Committees," are independent political entities that can raise and spend unlimited amounts of money to support or oppose political candidates and issues. Unlike traditional PACs, they can't directly coordinate with candidates. Super PACs play a significant role in modern election campaigns, shaping political landscapes through their financial influence. This thesis needs to take into account a lot of elements because, in the electoral process, it is necessary to evaluate all the possible elements or almost those that could have the larger effect and, therefore, could influence the outcome of the elections. Also if analysts disagree on the amount of the effect provided by the money on the results of the races, about all of them say that the volume of expenditures provided by each candidate has the possibility of helping them in getting elected. As expressed by Silberman and Yochum (1978), "*in the case of an election, the outcome (win/lose) can be viewed as dependent upon various inputs*" (p. 671). And, in the case of this thesis, the input chosen will be the expenditures provided by the independent entities but, as was said, a lot more elements will be considered because not only the possibility of spending more money than the opponent allows to win the election, otherwise Michael Bloomberg would have won the Democratic primary and the presidential elections in 2020, thanks to the fact that the billionaire has spent more than anyone else during the democratic presidential primary of 2020¹⁷. This work will try to apply the more possible elements in order to deepen a subject already extensively analyzed in the last decade.

The literature on the relationship between money and politics is vast and encompasses numerous research papers because, over the past years, political scientists have dedicated substantial effort to assess this topic, with a particular emphasis on the remarkable surge in total expenditures observed during the recent

¹⁷ (Henney 2020, <https://www.foxbusiness.com/money/boosted-by-bloomberg-2020-ad-spending-topples-record-breaking-1b>)

electoral cycles in the United States. Researchers have delved into understanding both the expenditures made by candidates' committees and the contributions from outside committees during general elections. Since the fact that the first presidential election held after the Supreme Court judgment was in 2012, some papers have decided to center their work solely on the primary election that preceded those elections. It is the case of the analysis made by Christenson and Smidt (2014), who observed that those committees *“contributed throughout the campaign, but their presence was particularly pointed during the nomination contest, where the demands for money are high but candidates are constrained in their ability to raise funds”* (p. 411). But, as was said, this was the first time that independent actors appeared at such a high level, and, in fact, since 2012, their presence and influence have risen, especially because of the amount of money that they have received as contributions reach, every election, a new high record¹⁸. During that primary election, as noted by many studies, the contributions of the Super PACs influenced the ability of Mitt Romney to win the Republican nomination for the presidency. As noted by Gulati (2012), in that contest the Super PACs linked to the former Governor of Massachusetts spent more than forty-six million dollars of which more than 85% with the aim of opposing the other candidates in the primaries of his own party. The expenditures made during the primary elections allowed Romney to win the contest, but not to win the general election against sitting President Obama, whose related Super PACs, as pointed out by the author himself, spent almost six million dollars in opposition to the candidates of the other party. Those numbers should show off the incredible importance that the Super PACs and other independent entities have, especially if it is considered the fact that those were the first presidential elections that saw the presence of this new and incredibly powerful tool. The primary system is an important element in the United States political landscape because this instrument allows citizens to choose their candidate by following strict rules that are different between states because some of them require party registration, while others do not permit the use of the voting by mail system. But the presence of independent expenditures has created a distortion inside it, especially in the states where the law requires party registration in order to vote in

¹⁸ (Federal Election Commission, https://www.fec.gov/files/bulk-downloads/2020/independent_expenditure_2020.csv)

the primary. In fact, in order to appeal to the voters of his or her party, each candidate will use two different methods: he or she will portray the other candidates as unfit for office, effectively discrediting him in the eyes of voters even in the event of his or her victory, or it will use increasingly divisive policies clearly aimed at its political alignment, polarizing a system that is already polarized given the mere presence of two nationally relevant political parties. In fact, according to some studies, the influence of independent entities on the primary system has contributed to the polarization of the U.S. political landscape because, as noted by Mann and Corrado (2014), *“Ideologically-based outside groups financed by wealthy donors appear to be sharpening partisan differences and resisting efforts to forge agreement across parties”* (p. 1). During the election period, which in the United States is almost constant given the continuous turnover at the local and federal level of elected representatives, citizens are daily bombarded with election advertisements that are helping to shape American public opinion. Furthermore, as expressed by Bassetti, Pavesi, and Scotti (2020), since 2010 independent expenditures have influenced the electoral system and the possibility for companies and unions to collect money for potentially anonymous independent expenditures has increased candidates’ polarization. Especially the negative advertising has allowed an incredible rise in the division between the two parties portrayed also by the fact that there are many fewer then-candidates and members of Congress considered bipartisan legislators, not just by the media, but also by the DW-Nominate index¹⁹. Thanks to all this information, it is possible to say that the importance of outside expenditures starts during the primary but continues during the electoral process. In fact, independent organizations receive daily contributions from the whole nation and spend at the same speed by filling the advertising markets, both television and online, with all kinds of content in order to spread their ideas and promote their candidates for each elected office. Therefore, the analysis, moreover if only elections to the House of Representatives, cannot be limited to an observation of the party primaries, but must go further, that is, it must analyze the behavior of outside expenditures during the general elections in November, in this specific case of November 2020.

¹⁹ (DW-Nominate, <https://www.voteview.com/congress/house>)

The only element that must necessarily be taken into account is the presence of some different legislation in the methods of implementation of the primaries between the various states. In fact, some states implement different primary modes that cause them to lead to a different selection of candidates in the general election. These differences, present in only a few states, play an important role in the outcome of the general election as they determine who will be the candidates in November. As Hirano and Snyder (2019) write in their book on primary elections in the United States, the substantial differences in the modalities decided by the states can only be observed in three states, namely California, Washington, and Louisiana. These differences are essentially based on the method and timing of candidate selection. The electoral systems of California and Washington state are quite similar and have been the subject of numerous debates and subsequent attempts at emulation, as in Montana for the 2024 election²⁰. The California system called Top Two Primary provides a single primary election for candidates of all parties where the two candidates who reach the highest number of votes advance to the general election where only they will be present on the ballot. Similarly, Washington state has a system called Nonpartisan blanket primary in which candidates from all parties compete in the primaries and only the two most voted will advance in the general election the following fall. This system was implemented in the last few years as an attempt to increase competition, reduce partisanship, and provide voters with more choices. One of the main arguments in favor of this type of primaries is that it encourages a more moderate and inclusive political landscape. Supporters argue that by eliminating the dominance of party politics, candidates are compelled to appeal to a broader range of voters, including those from opposing parties. This, in theory, could lead to increased bipartisanship and collaboration in government, as elected officials are incentivized to represent a diverse set of constituents. Critics, however, argue that this system can have unintended consequences. One concern is that it may marginalize minor parties and limit the diversity of voices in the general election. With only the top two candidates advancing, third-party or independent candidates may struggle to secure a spot in the final round, potentially leading to a lack of representation for certain ideological viewpoints.

²⁰ (McCormack 2023, <https://www.nationalreview.com/corner/montana-legislature-tables-bill-to-make-senate-election-a-top-two-primary/>)

Another criticism is that the system may favor incumbents and wealthy candidates who can afford expensive campaigns. Without the backing of a major party, lesser-known or underfunded candidates may face significant challenges in gaining visibility and support. In these two cases, voting in primary elections takes place well before general elections so that only the two most voted candidates are on the ballot papers in the general election. Louisiana proposes a similar model but with some differences. Its electoral system is unique and distinct due to its implementation of the top-two primary, also known as the "jungle primary." This system differs from the traditional primary system used in many other states, where voters choose a candidate from their preferred political party. In the Louisiana top-two primary, all candidates, regardless of party affiliation, run on a single primary ballot on general election day, and if a candidate gains more than half of the votes he or she will be elected, otherwise the top two will advance to a runoff. Despite these criticisms, the Louisiana top-two primary has been in place since the 1970s and has shaped the state's electoral landscape. It has contributed to more competitive elections, as candidates are compelled to build broader coalitions and appeal to a wider electorate. The system has also fostered a sense of political pragmatism, as candidates may need to appeal to voters beyond their party's traditional base. Overall, the Louisiana top-two primary system represents a departure from the conventional primary systems used in many other states. Its impact on fostering competitive elections and diminishing the influence of party politics continues to shape the state's political landscape. Ongoing debates surrounding its effectiveness and potential modifications underscore the ongoing evolution of Louisiana's electoral system.

As for the primary, there are many scholars who have observed the course of the general election and the behavior that those entities have held during these periods. In fact, as noted in the book of Franz (2013), since 2010 when the Roberts Court struck down the Bipartisan Campaign Reform Act (BRCA), States have been inundated with election advertisements so much so that

In the fall of 2012, voters in Wisconsin [...] were exposed to over 49,000 ads on broadcast television. Of those ads, 31%—over 15,000—were sponsored by interest groups. [...] Wisconsin was not unique, however. Across the country, voters in states with competitive Senate races were exposed to a

deluge of spending from interest groups. This included Montana, with over 70,000 ads in the fall re-election campaign of Democrat Jon Tester, 24% of which were from outside groups; Indiana, where over 40% of 40,000 ads were from groups; and Virginia, where 35% of over 39,000 ads were group sponsored. (p. 152)

Those numbers, as in the case of the Republican primary, refer to the first year in which a Presidential election was held and demonstrate how the independent expenditures had already shown their firepower during an election. Primary and general elections can be considered as a continuum, because once the first is finished, immediately after the second begins, and the presence of the subject of this thesis shows no signs of decreasing as anyone can see by looking at the donations received, and the expenditures made on the FEC online portal. As expressed by Meirowitz (2005), candidates take into account the continuum because they use primary to appeal voters and to formulate their electoral message so that they can reach as many citizens as possible. For this reason, some researchers, have examined both the general and primary elections considering them as interconnected, because expenditures and other campaign activities during the primary stage influence the outcomes in the general election. As noted by Cox (2022), *“Super PAC spending reveals their potential impact; [they] have been a major force in general elections and dominate primary spending by non-candidate committees”* (p. 2). As was said previously, the choice of observing both primary and general elections was taken by some authors because the influence of the independent expenditures does not start at the end of the primary, but it is continuously placed during the elections. Through the use of expenditures, especially in advertising, those entities have the power to move thousands, if not millions, of U.S. citizens and bring them to the polling place to cast their preferences, as expressed by Ansolabehere, Ivengar, and Simon (1999), and their ability to move people give them incredible power, also if other researchers, like Vavreck (2007), doubt money have such this influence but do have taken into account the role made by the main subject of this thesis, namely the Super PACs and the 501(c) groups. For this reason, the role of the primary election could be fundamental for the outcome of the election mainly if the district of the race seems to be sure that it will vote for a particular party or candidate given previous election results. In this case, the role of the primary is crucial

because the winner of the primaries will tend to be the winner of the general election. And, in fact, not in any case, the results of the primary in the United States seem to be fundamental for the outcome of the general elections. On the other hand, Born (1981) said that until that moment it was unclear if the possibility for a candidate to win a district is negatively correlated with the presence of a competitive primary, also if party officials continue to express a lot of criticism for such a form of contest in which they have to waste a lot of money. His research discovers that the presence of a challenger for the incumbent during the primary does not decline the support for the candidate during the election of the following fall, especially if those challengers are mostly “amateurs” and represent an unserious threat. Years later, many other political researchers have tried to address such a question, obtaining the same results as those achieved by Born. As expressed by Lazarus (2005), the primary results impact the outcome of the general election only in some cases that are limited by the presence of two possible factors. This happens especially if the primaries are highly divisiveness for the party of the incumbent candidates, causing unintended consequences of the citizens’ behavior, and if the electoral district in which the candidates at the primary have decided to run seems to be more competitive for the party of the challenger, and this last factor will be addressed in the following pages. The results of Lazarus’s research show that if a candidate expects not to win in the November elections then there could be a competitive primary and these expectations tend to also mark the outcome of the challenges of the following autumn. At the same time, a competitive primary seems to damage more an incumbent candidate, which will do poorly, than a challenger one, which will do well in the fall. The question of district competitiveness seems to be quite more complicated because it is necessary to find a measuring instrument suitable for the purpose. This device could be represented by the Cook’s Partisan Voting Index²¹ which evaluates the partisan leaning of each state and college using the difference between the mean of the results of the last two presidential elections in each place with the national mean results of the same two competitions, giving a continuum variable that on the left has the most Democratic-leaning districts while on the right has those that are more Republican-leaning. According to the value expressed by this index, the districts that have a value between D+5 and R+5 are those

²¹ (The Cook Political Report, <https://www.cookpolitical.com/cook-pvi>)

more competitive. Those two elements, namely the competitive primary, and constituency, will be central in the analysis provided by this thesis because they allow, according to Rocca and Clay (2021), an understanding of how the committees of the candidates decide to allocate resources, but also how independent entities act in order to promote their principle, trying to elect those candidates that they prefer. The system of the primaries gives more than one problem for the purpose of this thesis: it forces to consider a lot of possible control variables causing an implementation of the work that could be simply avoided by considering the two just mentioned elements inside the analysis of the general election. By using the incumbency status of the candidates and the partisan leaning of each district, it is possible to exclude the presence of the primary election inside the analysis, as was made by Ferguson, Jorgensen, and Chen (T. Ferguson, Jorgensen, and Chen 2019) in their analysis of how money drives the outcome of the general election.

Therefore, before setting out the reasons that led to the decision that led this thesis to just consider the outcome of the general elections without observing the results of the primaries, it is necessary to remark on the importance of incumbency status and district competitiveness in the whole United States political system. A lot of studies explore the complicated relationship between incumbents and their probability of winning another term in the House of Representatives. According to Levitt and Wolfram (1997), since 1997 and “*over the last two decades, well over 90% of the incumbents seeking reelections have been successful*” (p. 45) and this advantage has continued to grow for many years also if no one has identified a clear element that allows to be defined as clear winner of giving them another term in the House. Cox and Katz (1996) have tried to expand the knowledge about this topic observing that many authors in the past years have used as independent variables various resources like opportunities to perform constituency services and the partisan dealignment. At the end of their research, they noticed that “*it is crucial to understand why having had previous electoral experience became more and more important in predicting vote shares*” (p. 479). But after some years, Jacobson (2015) observed that such advantage was declining in conjunction with an upsurge in party loyalty, the prevalence of straight-ticket voting, and the growing phenomenon of president-centered electoral nationalization, these developments are outcomes of the deepening and increasingly

consolidated partisan divisions among the American electorate. Looking at the results of the House of Representatives elections since the election of 1994, when the Republican party regained the majority for the first time in about forty years, it is possible to note that, apart from in three cases (the elections of 2006, 2010 and 2018), the number of districts that changed parties was very low, causing the same representatives to remain for many years, as can be seen with the presence of some representatives who were sworn in before President Reagan took office. Therefore, the incumbents have, at least in theory, a general advantage due to the fact that their name is better known within the district, and they have woven a long-lasting network of both public and private relations during their tenure in Congress. According to the research conducted by King and Gelamn (1991), it is possible to notice that through the removal of incumbency advantage and a careful analytical examination of the existing configuration of incumbents and challengers, the analysis uncovers an underlying electoral system that consistently exhibits a bias in favor of the Republican party. These findings challenge the conventional wisdom by revealing that while incumbency advantage does impact the electoral system, its effect on partisan bias is more substantial than its influence on electoral responsiveness. The results suggest that the prevailing partisan bias remains intact despite variations in electoral dynamics, highlighting the need for a deeper understanding of the factors contributing to this enduring bias. Moreover, another study made by Jacobson (1978) notes that the incumbency variable has highlighted a noteworthy and positive relationship between the amount of money spent by challengers (i.e., candidates who do not currently hold the seats) and their probability of winning the general elections in the districts in which they are running. Conversely, the relationship between incumbent candidates and their expenditures tends to be more complex, often displaying little to no effect on the electoral outcome. Furthermore, models indicate that the incremental benefits resulting from increased campaign spending are considerably higher for challengers than for incumbents. This clear-cut observation leads to the inference that any increase in spending by both candidates will confer an advantage to the challenger. However, it is important to note that the focus of this paper primarily lies on the expenditures provided by the candidates' committees, while external expenditures are not taken into account mainly because when the research was conducted it was even impossible to

foresee what the consequences might be from the judgment pronounced more than forty years later by the Roberts Court. In the paper made by Stone, Fulton, Maestas, and Maisel (2010), the analysis of the topic goes even deeper, looking at the attributes and personal qualities of the candidates to help explain why incumbents enjoy sizeable electoral advantages. But, looking at the results of the 2020 primaries for the House of Representatives in all the U.S. states, just eight incumbents have lost their primary election to a challenger (Benen 2020), each of whom was going to win the general election in the following fall. As it is possible to notice in the electoral information provided by major newspapers and television networks, each incumbent that has lost the primary has decided to endorse the winner of the race in order to help him or her in being elected. Such a result allows to avoid the problem caused by the divisiveness of primaries, giving the possibility of treating the district as if it is a seat held by the incumbent party. Through the means used by the articles just mentioned it will be possible to observe the result of the elections taking into account the result of the candidate or the incumbent party, also checking it through when spent both by the challenger and by independent organizations to support the latter.

As was said previously, another important element that is necessary to take into account is the strict relationship between the competitiveness of the district and the amount of expenditures made in each college by candidates, parties, and outside committees. In a study conducted by Huckfeldt, Carmines, Mondak, and Zeemering (2007) it is possible to take into account the connection between the competitiveness of the electoral college and the turnout at the ballot but in their conclusion, they said that *“the machinery of elections to the House of Representatives produces a net bias against political competitiveness”* (p. 818) because the electoral competitiveness of the districts does not lead an increase in voters’ turnout. Like the other authors, Jordan (2017) has tried to observe which elements influence voter turnout, including campaign expenditures. In fact, the results of the analysis show that the total campaign expenditures always keep a positive and high significant correlation with the turnout amount, meaning that the higher the expenditures made by all the actors, the higher will be the number of people that will go vote at the poll stations. So, it is possible to say that expenditures influence the probability that a citizen will go vote is crucial in the electoral process because just by using any amount of money, they have the

possibility of influencing millions of voters. Consequently, the fact that the perception of living in a more or less competitive district does not change the opinion of citizens on going to vote could be very relevant, as it indicates that there would be no major variation in spending between a swing district and a strongly Republican or Democratic one. Taking into account that element, it is necessary to evaluate the relationship between expenditures made by all types of committees and the partisan leaning of each district. Thanks to previous research that assesses the connection between expenditures and voter turnout, it is possible to hypothesize that in the most fluctuating districts the candidates, and with them the parties and other external actors, are more likely to spend in order to incentivize people to go and vote for them. In this way they increase their spending and also force opponents to increase the budget for the elections, greatly increasing the cost in these constituencies. Just to make an example, as expressed in the book of Schatzinger and Martin (2020), the Super PAC linked to Trump's first presidential campaign, only on election day, has spent about three hundred million dollars in most swing states, like Wisconsin, Michigan, and Pennsylvania. Surely not only thanks to this immense last-minute expense, Trump managed to win the election by becoming President thanks to the electoral college's votes of the most contested states. Looking at a single electoral district, it is possible to notice that the situation seems to be the same because most competitive districts are covered with money coming not only from candidates in the district but also from parties, other candidates, and independent entities. As described by Fossen, Schweidel, and Lewis (2019), strong partisan leaning implies less fundraising for the candidates but also could reduce the need for more advertising in the district during the elections if it is considered to be very one-sided. Moreover, studies consistently reveal that candidates in districts with a more competitive PVI tend to allocate higher campaign expenditures in an effort to secure victory. The PVI serves as a significant indicator of the strategic allocation of financial resources in electoral campaigns. The district's partisan leaning can be examined through the process of redistricting. By exploring campaign spending on fundraising activities and advertising and considering the shift towards more competitive Cook PVI categories, Quigley's (2019) research delves into the adaptation of campaign spending strategies in response to newly established congressional districts. This approach allows for a deeper understanding of how

candidates navigate and allocate resources in light of evolving district dynamics. These two presented variables will be used in the analysis as control variables for the research. They could allow a better understanding of the topic, because they make it possible to verify that the incumbency of the seat and the competitiveness of the district can be two of the factors, if not the only ones, to most influence the election result. Like Nielson and Visalvanich (2017), the thesis will adopt these variables as the control for the outcome of the dependent variables, which is the outcome of the general elections. Each of them will be thoroughly observed and analyzed in Chapter 2, where they will be related to the expenditure made by both candidates and outside actors. In that chapter will be possible to observe the distribution of all the main variables that are described in this one, especially the PVI and the expenditures made by both the candidates and the independent actors.

As mentioned above, the analysis conducted in this thesis will not take into account the first electoral phase held in the first half of 2020 but will focus only on the results of the November elections. As already experienced by many analysts, it is possible to take into account only these elections if anyone takes some precautions, especially those set out in the previous two paragraphs. In fact, many of the studies that analyze this same subject avoiding primaries, such as that of Ferguson, Jorgensen, and Chen (T. Ferguson, Jorgensen, and Chen 2019), take into account as control variables the two previously mentioned, namely incumbent status and the partisan leaning of the district. This thesis will employ the same method adding some other possible control variables which will be mentioned on the following pages. Besides, using this type of approach allows to avert another possible complication of analysis because it allows not to take into account the rules for the primaries decided by the states for the selection of candidates for the following autumn. As expressed many times, according to a lot of studies there is a strict connection between campaign spending and electoral outcome. According to the analysis mentioned above, *“the relations between money and votes cast for major parties in elections for the U.S. Senate and House of Representatives from 1980 to 2018 are well approximated by straight lines”* (p. 527), also if it is possible to notice, and the authors explain, that there three cases in which the general rule has not been respected (1980, 1996, and 2012). Once more, in this study, the main independent variable, the amount of money

spent, seems to be related to the outcome of the elections. Money has a central role in the actual political system in the United States, especially because, as expressed by Gerber (1998) in his work on campaign spending, *“if campaign spending matters for election outcomes, it is a short step to conclude that campaign finance has a significant influence on who runs, who wins elections, and what the winners do in office”* (p. 403). Many studies have concluded that the amount of money that candidates spend during the voting for the renewal of the entire House of Representatives can direct the election result in favor of the candidate who spends most on the challenge, although exceptions to this trend remain, as anyone can read in the previous pages. But in most of those same analyses, there is more talk of the expenses made by the candidate and his own committee, the external contribution guaranteed by the constant presence of the outside expenditures is not evaluated. Since 2010 it should have been necessary to evaluate the contribution that these special Political Action Committees have had within the already complicated U.S. electoral system. On the contrary, it was preferred to take into account many other factors, including some that will be used in the analysis of this thesis, rather than going to deepen an issue that could be really complicated. The use of general elections only, as done in Paul and McClellan's (2022) analysis, should allow to answer the question “How was outside spending helping each candidate in the House of Representatives races?”. Those two authors have tried to interpret the role of outside spending during the Georgia Senate elections of 2020, the two races that give back to the Democratic party the control of the upper chamber of Congress. During that election cycle, independent expenditures erased the previous spending record by distributing more than two billion dollars across the country. As noted by Baker (2018), those entities have rightfully entered among the largest contributors to the election spending system thanks to the fact that they can raise and spend funds without any legal limit as long as they keep their independence from candidates and political parties. Thanks to the fact that those organizations are now major spenders in the House races, It is necessary to observe whether their strategic contribution in terms of the amount of expenditure can be fundamental in supporting or opposing a particular candidate, specifically if their opposition to a candidate helps the opponent to win the elections. Considering these factors, one could assume, as Farrar-Myers and Skinner (2013) did, that the independent expenditures implement strategies that can be

functional to the election of one or the other candidate, acting as allies of their campaigns, spreading the message transmitted or reproducing it with a greatly increased frequency. Through the implementation of a sort of parallel campaigns, those actors act as "amplifiers" of the candidates without being able to coordinate with their campaigns, as required by law.

Given that until now this work has talked about independent or outside expenditure, sometimes also by means of naming some of the actors in the field, it is now necessary to explore the classification of these entities by observing similarities and differences and explaining their importance within the U.S. system. According to the FEC "*individuals, groups, corporations, labor organizations, and political committees [...] may support or oppose candidates by making independent expenditures. Independent expenditures are not contributions and are not subject to limits*²²". The two main entities that play this role are the Super PACs and the 501(c) groups, better known as dark-money groups. There are similarities between them but there are still some substantial basic differences that characterize them. The definition provided by the Federal Election Commission website says that the Super PAC is

*a political committee that makes only independent expenditures that may solicit and accept unlimited contributions from individuals, corporations, labor organizations, and other political committees. It may not accept contributions from foreign nationals, federal contractors, national banks, or federally chartered corporations. Such committees, also known as independent expenditure-only committees, must register with the Commission and comply with all applicable reporting requirements of the Act.*²³

Thus, Super PACs collect and spend unlimited money from any type of donor subject only to certain constraints imposed by FEC regulation. This legislation is specific only to this type of committee and therefore a specification is necessarily required. Although the constraints envisaged for the Super PACs are not very stringent, there are some committees that must be subject to even less stringent legislation. These are the 501(c) groups, also known as dark money groups, which have the only

²² (Federal Election Commission, <https://www.fec.gov/help-candidates-and-committees/making-independent-expenditures/#IE>)

²³ (Federal Election Commission, <https://www.fec.gov/help-candidates-and-committees/filing-pac-reports/registering-super-pac/>)

constraint that they cannot spend most of their budget to finance politics. As an article by Melissa Yeager²⁴ explains, these groups do not have to show their donor lists and are not bound by the control of the electoral commission. It is often difficult to find out their activities mainly because they do not have to declare how they use the funds to the FEC and sometimes the only way to do it is to look at the reports that the Federal Communication Commission (FCC) obliges advertising transmitters to publish online, although there is no standardization of the type of information they have to enter and the reports they have to compile for the FEC tend to be published much later than those of the Super PACs. In addition, these entities can act as intermediaries between donors and Super PACs since the fact of not having to show the list of their funders guarantees the anonymity of those who decide to give large sums of money in support of one or the other political force. For the purpose of this thesis, all the entities that made independent expenditures with the purpose of trying to elect or avoid the election of a certain candidate will be taken into account to observe their role during the elections.

Each of these entities plays a fundamental role during the electoral period and this thesis will observe it, as has been repeatedly said, only during the period between the end of the primaries and the day of the general elections, which in this case is November 3, 2020. The extensive literature on money and politics provides valuable insights into the influence of campaign expenditures. Researchers have employed diverse approaches, models, and data sources to investigate the relationship between spending and electoral outcomes. Through these endeavors, they have shed light on the complex dynamics at play, emphasizing the need for a reevaluation of the theoretical assumptions regarding incumbent campaign spending and its impact on vote shares. As a result of these collective research efforts, a significant portion of the authors concur on the necessity to reassess the theoretical correlation between incumbent campaign spending and incumbent vote shares. It is proposed that if incumbents prioritize maximizing their probability of victory rather than increasing their votes, traditional incumbent campaigns will yield only marginal gains in terms of share at the polls. However, there is an emerging consensus that the results indicate that independent expenditures contribute to a slight increase in overall campaign

²⁴ (Yeager 2015, <https://sunlightfoundation.com/2015/10/30/the-difference-between-super-pacs-and-dark-money-groups/>)

spending and tend to benefit Republicans in general elections, albeit with substantial variations across different scenarios. In summary, the extensive literature on money and politics provides valuable insights into the influence of campaign expenditures, also because after the judgment pronounced by the Supreme Court it was clear to everyone that the electoral system would necessarily change, indeed it would even be distorted by the entry of these new actors. In fact, as has also been noted in the study made by Magleby (2014), since 2010 many analyzes have gone in the direction of hypothesizing how this distortion of the system has caused a disproportionate advantage for the Grand Old Party (GOP) especially because, at least according to polls and electoral exit polls²⁵, citizens with a higher income tend to vote for the elephant party. As highlighted in Benton's (2012) study, the surge in campaign expenditures during general elections in recent years comes as no surprise. Looking back to 1952, a presidential election year, the combined cost of political campaigns at both the national and state levels amounted to \$140 million. Fast forward to 2000, and that figure skyrocketed to \$3.1 billion. By 2008, estimates reached a staggering \$5.3 billion. In the 1960s, it was not uncommon for a successful House candidate to spend less than \$100,000. However, by 2008, the average cost of winning a seat in the U.S. House of Representatives had surpassed \$1.2 million. Similarly, winning a U.S. Senate seat in the 1960s could still be achieved with a budget of less than half a million dollars. In 2008, the average Senate race cost nearly \$9 million. These escalating figures underscore the tremendous growth in campaign expenditures over the years and the increasing financial demands of political campaigns, reflecting the evolving nature of modern elections and the intensifying competition for political office. This incredible increase in general election spending was not only caused by the appearance of organizations outside the countryside, although the largest increase is subsequent to the Supreme Court ruling. Moreover, this considerable increase in campaign spending has also had important consequences in another area, that of advertising. Just to give an example, evaluating the spending on advertising made by all campaigns in the year of the electoral challenge between Obama and McCain in 2008 (\$ 1063616156)²⁶ and the spending of the candidates in the year of the challenge between Trump and Biden

²⁵ (Presidential exit poll, <https://www.washingtonpost.com/elections/interactive/2020/exit-polls/presidential-election-exit-polls/>)

²⁶ (OpenSecrets, <https://www.opensecrets.org/campaign-expenditures?cycle=2008>)

in 2020 (\$ 8550669935)²⁷ it is possible to note an incredible increase of more than seven hundred percent, as shown by the data provided by OpenSecrets. This increase is largely due to the introduction of independent spending within election campaigns, but some research, such as the one made by Cox (2022), has noted that *[independent expenditures] may not only have influenced who won the election but also candidate positions*, expressing the possibilities that the previously mentioned influence in the elections provided by those outside organizations may have moved the candidates far from the ideological center to the extreme positions of the political spectrum. leading to a consequent increase in the polarization of the political system. This extreme division is partly due to the amount of mostly negative information that television, radio, social media, and billboards cover the entire country for several months. Therefore, it is possible to say that expenditures, especially those made by outside organizations, against a candidate are responsible for this further polarization. Especially the aforementioned first year of presidential elections after Citizen United represented a flood of negative publicity against challengers in both the primary elections and the elections of the following November. As can be seen in the study by Fowler and Ridout (2013), since the first elections that saw the presence of independent expenses the majority of the advertisements purchased (about 75%) had a negative sentiment like the predominant one.

The increasing presence of clearly negative advertisements during election campaigns has led to numerous analyses being conducted on them. Moreover, taking into account the data provided by the FEC, it can be seen that the Commission employs a classification system that categorizes the money spent by independent organizations based on their intended purpose, creating a dichotomous variable that indicates whether the expenditure was made to support or oppose a specific candidate. The use of advertising to either support or oppose a candidate profoundly influences the type of information conveyed to citizens. When an independent organization decides to support a particular candidate, it will strive to transmit positive information about the individual they endorse. Conversely, if their intention is to oppose another candidate, they will employ different messages characterized by negativity in an attempt to dissuade citizens from voting for that particular candidate. This distinction in

²⁷ (OpenSecrets, <https://www.opensecrets.org/campaign-expenditures?cycle=2020>)

messaging strategies highlights the significant role of outside committees in shaping the information landscape and influencing voters' perceptions during electoral campaigns. Moreover, the impact of positive or negative messages disseminated within a particular district can extend beyond its borders, influencing neighboring constituencies as well. Spenkuch and Toniatti's (2016) analysis tries to estimate the persuasive effects of political advertising, aiming to challenge previous studies suggesting that political ads have minimal or negligible impact on voter choices. Given that a substantial portion of independent expenditures is dedicated to TV advertising, this dynamic interplay among citizens, outside organizations, and the classification of expenditures as supportive or opposing becomes pivotal for the objectives of this work. As mentioned above, the increasing distribution of mostly negative political messages has accelerated a process of progressive polarization of the two camps, with the Republican Party positioning itself increasingly to the right and the Democratic increasingly to the left. As a study by Lau, Andersen, Ditonto, Kleinberg, and Redlawsk (2017) shows, increasing exposure to advertising with a clear negative message is considered one of the main elements of polarization in the political system. Moreover, considering that the use of a negative connotation in electoral messages would have the power to energize citizens by pushing them to go to the polls, even those who vote less frequently. Looking more closely at the data relating to the distribution of advertisements with negative messages, as was done by Dowling and Krupnikov (2016), it can be seen that the system has gone from 1960 when these accounted for only 10% of the total messages transmitted, while just over fifty years later, in 2012, the advertisements that had a more positive content had become only 14% of the total. In addition to all the positive effects of negative advertisements, political analysts have always taken into account that negativity in political campaigns can indeed diminish voters' perception of the targeted politician. However, it is important to note that negative campaigning can sometimes result in a backlash effect for the attacker, as noted in the study proposed by Galasso, Nannicini, and Nunnari (2023). While negative messaging may succeed in tarnishing the image of the opponent, it can also elicit a negative response from voters who perceive the attacker as engaging in unfair or unethical tactics. This backlash effect occurs when voters sympathize with the targeted politician or perceive the negative campaign as an

attempt to divert attention from substantive issues. Consequently, negative campaigning can have unintended consequences, potentially mobilizing support for the candidate being attacked rather than achieving the desired outcome for the attacker. Political strategists and candidates must carefully consider the potential backlash when deploying negative campaign strategies, weighing the potential benefits against the risks of alienating voters and damaging their own reputation in the process. On the contrary, the minority share of independent expenses is made in support of candidates. As Brader (2005) notes in his work on the motivations that drive voters, while the ads in opposition tend not to please one of the candidates, those in support try to push for the candidate in question to be appreciated by voters, through a more confident and hopeful message. Another substantial difference highlighted by Weinberger and Westen's (2008) work is that more negative election messages have a greater impact on the possibility of citizens going to the polls and, for this reason, campaign managers tend to use them more. In any case, the presence of a significant share of advertisements bought by independent organizations containing positive messages must be taken into account within the analysis, since it influences the perception that voting citizens have of candidates for the lower house of the U.S. Congress. The choice of which messages to convey and which types of emotions to use is up to analysts and consultants of electoral campaigns (or independent organizations more specifically in this case), who through focus groups, surveys, and other types of analysis will prefer a certain type of electoral communication. Taking into account the classification proposed by the FEC, it can be seen that the expenses made by independent organizations always benefit one or the other candidate, since, for example, an expenditure made against the Democratic candidate will favor the Republican one and vice versa. Messages of support combined with those in opposition to the other candidate tend to favor the same person, always with the goal that independent spending is set at the beginning of each election campaign. The sum of spending on support of Democratic candidates and the one in opposition to Republican candidates, and vice versa for the other party, will be crucial in determining the votes obtained by the two campaigns in the November 2020 general election. Then, the classification provided by the FEC within the dataframe used for the analysis and better presented

in the following chapter will be used to define which candidate will benefit from each expenditure made by entities independent of the campaigns.

So, as anyone can well understand, the goal of every dollar spent by both election committees and independent organizations is to win a seat or to have it go to a particular candidate. Each of the elements already mentioned or those that still need to be read could influence the number of votes that both candidates will receive in the various districts. Most policy analysts and researchers use it as a measure of the possible increase or decrease in the number of votes, not their total number but percentage. As in the analysis of Berry, Berkman, and Schneiderman (2000), the one made by Krebs (1998), and that of Hogan (2004), this work will also use the percentage of votes as an independent variable of the research and this decision is subject to three fundamental points. First, vote share is undeniably a more accurate measure of a candidate's popularity than simply focusing on the number of votes won, but also on the number of seats won. While seat counts indicate success in specific constituencies, they fail to reflect the overall public sentiment. Vote share, on the other hand, reveals the true extent of support garnered by a candidate across the entire electorate. It considers every vote cast, making it a fairer representation of people's choices. This metric enables a deeper understanding of a candidate's appeal, and their ability to resonate with diverse demographics, and provides valuable insights for future campaigns. In essence, vote share is an indispensable tool for assessing true popularity in a democratic process. Second, vote share is a vital metric that allows for meaningful comparisons across various elections, as can be seen in the study of Fair (2007). Unlike absolute vote counts, which can be influenced by population size and voter turnout, vote share represents the proportion of votes a political party or candidate receives relative to the total votes cast. This makes it a fairer and more comparable measure when analyzing election outcomes across different regions. By focusing on the percentage of votes, rather than raw numbers, it is possible to better understand trends, shifts in public sentiment, and the popularity of parties or individuals in a standardized manner. Ultimately, vote share provides valuable insights into the democratic process and political landscape. And third because in the realm of elections and polling, vote share becomes more likely to achieve statistical significance as the size of the sample increases. Statistical significance indicates the probability that observed results are not

merely due to random chance, but rather reflect a genuine trend or pattern in the larger population. With a larger sample size, the margin of error diminishes, providing greater confidence in the accuracy and reliability of the vote share data. As researchers and analysts strive for robust and meaningful insights, they emphasize the importance of larger, more representative samples to draw valid conclusions and make informed decisions in the democratic process.

Moreover, the analysis of election results in districts, especially in the case of the United States general election, is subject to the need to observe an additional control variable that may have had at least a minimal effect, especially as a result of the polarization and nationalization of electoral campaigns. As Burden and Kimball (1998) note, the simultaneous presence of other elections for an office with a wider scope than that of a single district, whether presidential, gubernatorial, or senatorial, could be fundamental in obtaining a certain type of election result rather than another. This simultaneous election scenario limits the opportunity for voters to engage in ticket-splitting, as they may face constraints in their ability to vote for candidates from different parties across different offices. Consequently, the presence of these concurrent elections further emphasizes the significance of understanding the dynamics of this element and its impact on electoral outcomes. It highlights the complexities faced by voters and the potential influence of factors such as party loyalty, electoral strategies, and the alignment of political races in shaping voting behavior. Ticket-splitting refers to the practice of voters casting their ballots for candidates from different political parties in the same election. Rather than consistently voting for a single party's candidates across all offices (such as voting for all Democratic or all Republican candidates), ticket-splitting occurs when voters choose candidates from different parties for distinct positions. For example, a voter might cast a ballot for a Democratic candidate for President but opt for a Republican candidate for the House of Representatives or the Senate. Ticket-splitting can indicate a level of political independence or a willingness to evaluate candidates individually rather than strictly adhering to party affiliations. It allows voters to mix and match their choices based on their assessment of each candidate's qualifications, policy positions, or personal appeal. As can be noted just by looking at the electoral results since the mid-twenty century, as noted by Campbell and Miller (1957), there has been a constant

presence of this phenomenon, especially in view of the fact that presidential elections have been won mostly by Republican candidates, while congressional elections, both in the Senate and in the House of Representatives, were almost always won almost always by candidates of the Democratic Party that have held control of the legislative branch of U.S. power almost continuously for fifty years. In fact, as expressed by Beck, Baum, Clausen, and Smith (1992), "*the primary source of divided government in the United States is voters who split their ballots between the parties*" (p. 917). On the other hand, since the 1994 elections with the return of the Republicans in majority to the House of Representatives, there has been a trend contrary to the previous one, especially since from that moment the general inclination of the population was to vote for the same party regardless of the role that the candidate would play. Ticket-splitting can be influenced by factors such as the candidates' personalities, campaign messages, or local issues. Ticket-splitting can have significant implications for electoral outcomes as it can lead to a divided government, where different parties control different branches of government or legislative bodies. It also highlights the complexity and diversity of voter preferences within the electorate. Due to a prevalent tendency among citizens to vote for candidates primarily based on party affiliation, the occurrence of ticket-splitting appears to be relatively limited. Consequently, it is plausible that a candidate running for a higher office, such as President, may attract more votes for candidates from their own party in state or district congressional races, solely due to party association, even if voters may not possess extensive knowledge about these particular candidates. This phenomenon can be attributed to a form of party loyalty and the inclination of voters to support candidates from their preferred party across multiple levels of government. The impact of party affiliation on down-ballot races highlights the influence of partisan identity and the potential for coattail effects, wherein the success or popularity of a top-of-the-ticket candidate influences the outcomes of lower-level races. This underscores the importance of party branding and the challenges faced by candidates in educating and engaging voters about their individual qualifications and policy stances in order to stand out in a crowded electoral landscape. Therefore, as can be deduced, the influence of the results of the other major election campaigns within the district could influence the outcome of those in the House of Representatives, both from the same district and neighboring ones. Despite

being less frequently used, split-ticket voting continues to be present in the U.S. political system, Born (1994) has reviewed the model proposed by Fiorina in which she argues that the increase in split-ticket balloting in previous elections is due to at least a partial distancing of parties both from voters and from each other, making citizens prefer votes for candidates of different parties to balance the polarization.

But not only elections held at the same time play a crucial role in determining the victory or defeat of a candidate. As already mentioned above, the party tendency of a district tends to define numerous aspects, including above all the chances that a particular candidate can win the elections. As Krasno and Green's (1988) study notes, it often happens that a candidate judged to be of good quality for some of his personal characteristics decides to compete in one district instead of another. This decision is subject to certain factors, first of all, the real chances the potential candidate has of winning the elections. In 2022, for example, New York state districts were drawn to include incumbents from different boroughs within a single seat due to declining state constituencies. This led to the decision of Sean Patrick Maloney (former NY-18) to run in the seventeenth district he considered most secure and forced the incumbent of that district, Mondaire Jones, to seek re-election in the tenth. But in the end, neither of them managed to regain their seats in the House of Representatives. Similarly, analysis conducted by Bond, Covington, and Fleisher (1985) notes that previous election results affect a candidate's ability to be elected far more than his or her ability to raise money and political experience. In addition, other studies have observed the centrality of the role of the historical election results, noting some peculiarities that lead the U.S. to think that these do not only constitute the past but, on the contrary, play a crucial role in determining the outcome of subsequent elections. This is the case of the study conducted by Forsythe (1991) in which it is observed that the previous results influence both the historical context and the behavior of voters. Previous election outcomes offer valuable historical context and insights into long-term voter behavior. By analyzing past results, political analysts identify patterns and trends in voting preferences. They study party performance to strategize and allocate resources effectively. Electoral coalitions, formed by specific voter groups supporting parties, influence campaign strategies. Furthermore, voting shifts reveal changes in public sentiment over time, necessitating adaptability in party platforms. Historical context

enables parties to learn from past successes and failures, shaping their approaches for future campaigns and aligning with evolving voter concerns. Voter behavior holds crucial significance in elections, and past elections offer insights into decision-making processes. Analyzing previous data allows for understanding demographic influences on voting patterns, guiding targeted campaigning. Issue prioritization, as revealed by historical results, helps parties focus on relevant policies to resonate with the electorate. Swing voter analysis uncovers potential shifts in allegiance, impacting campaign strategies. Predictive models based on historical data aid in forecasting future election outcomes, considering historical vote shares, economic indicators, and incumbency status. Analyzing voter behavior helps anticipate trends and make informed decisions in the dynamic realm of politics.

Therefore, the history of the elections could influence the behavior of organizations outside the electoral campaigns that will buy more in those places where the candidate has the greatest chance of winning, making expenses of a strategic and non-ideological nature. An example of this could be the fact that Planned Parenthood Votes, one of the most prominent pro-Democratic candidate organizations, spent almost zero dollars in Hawaii while spending almost a million and a half to support Dem candidates in Michigan²⁸. The selected independent variables, as was said multiple time in the previous pages, is the independent expenditures made by outside organizations, like the Super PACs and the 501(c) groups. Those expenditures, as noted by Herrnson (2017), were mostly done in opposition to a candidate, and during general elections, avoiding contributing during primaries, runoffs, and special elections. But this last fact, notes the same work, also depends a lot on the affiliation of independent organizations: if they are affiliated with a party they will tend to invest even during the primaries, while if they are totally independent they will concentrate their spending more on the general elections. In contrast, mixed-interest groups tend to favor Republicans, with more than half of support spending in favor of them, as well as more than 90 percent of those in opposition, in the cycles analyzed by the author. As Heberlig and Larson (2014) note, the federal campaign system has evolved into two distinct systems, each with its own set of rules and regulations. The first system pertains to candidates running for federal offices, such as the presidency, the Senate,

²⁸ (Federal Election Commission, <https://www.fec.gov/data/committee/C00489799/?cycle=2020>)

and the House of Representatives. These candidates must comply with Federal Election Commission (FEC) regulations, which include contribution limits, disclosure requirements, and restrictions on the use of funds. The second system is characterized by the emergence of independent organizations and their expenditures, which operate independently from candidates and their campaigns. They have the possibility of raising and spending unlimited amounts of money, including funds from corporations, unions, and individuals. Unlike traditional campaign committees, outside organizations are not permitted to coordinate their activities directly with candidates. This division in the federal campaign system has led to significant changes in campaign finance dynamics. Candidates and their campaigns must navigate strict regulations and fundraising limitations, while independent entities have greater flexibility in their fundraising and spending activities. This has resulted in an increased level of spending by outside groups, often with significant influence on election outcomes. The presence of these two systems has sparked debates about the impact of money in politics, the potential for undue influence, and the role of transparency in campaign finance. Understanding the complexities of these dual systems is crucial for analyzing the evolving landscape of federal elections and the influence of money in shaping political outcomes.

As has been said many times, independent expenditure must be clearly distinguished from that made by election campaigns. These, unlike the former, are subject to much more stringent legislation, especially in terms of how much they can receive from a single individual. The presence of this type of expenditure is fundamental within the analysis, as it allows candidates to have funds to spend as they prefer, always with the aim of being elected. Through campaign money, candidates have the opportunity to influence citizens' attitudes toward certain policies, making them prefer some over others. As a study by Potters, Sloof, and van Winden (1997) shows, the campaign spending of both incumbent candidates and challengers allows voters to better understand which candidate might be better between the two proposed. In this, the funds of the candidates' electoral committees play a central role because, through the use of information, they influence the public perception of the two competitors in the general elections. Another study on elections related to gender differences by Atmor, Harsgor, and Kenig (2022), shows that election results are also

linked to the possibility of candidates being able to spend more than challengers. Greater spending by candidate committees, therefore, increases the chances of victory they have throughout their electoral journey. At the same time, the study shows that thanks to greater spending, candidates are able to erase the initial differences, in the case of gender, allowing them to conduct an au pair campaign regardless of their initial personal characteristics. Like the expenses made by independent organizations, those made by the candidates of the committees have also increased, just noticing that in 2008²⁹ the candidates for the House of Representatives spent a total of about nine hundred and fifty million dollars, while only twelve years later, in 2020³⁰ the year of analysis of this thesis, the candidates spent just under one billion eight hundred million dollars. In every type of U.S. election, as Moncrief's (1992) analysis suggests, there has been a sharp increase in campaign spending over time, although there have been some cases of little or no contest states where the cost of each vote obtained has remained roughly the same. Committee spending, as can be seen from this previous research, is central to all election campaigns. The analysis conducted by this thesis will necessarily have to take them into account in order to verify their influence on the electoral result using them as control variables within the model that will be used.

Numerous previous studies have shown that the relationship between money invested in elections and election outcomes suffers from a potential problem of endogeneity caused by the reciprocal causation of the dependent and independent variables that this thesis aims to use. As expressed by Green and Krasno (1988)

If incumbents raise money as they become threatened by the prospect of defeat, the apparent effect of incumbent expenditures will be biased positively, and the role of incumbent spending will be understated. Conversely, if challengers attract campaign contributions as their prospects for victory improve, the estimates for challenger expenditures will be biased positively, and the effect of challenger spending will be overstated. (p. 886)

Although most studies, such as Jacobson's (1978), focus on expenses made by candidate committees, expenses made by independent organizations are also subject

²⁹ (Federal Election Commission 2008, https://www.fec.gov/data/spending-bythenumbers/?office=H&election_year=2008)

³⁰ (Federal Election Commission 2020, https://www.fec.gov/data/spending-bythenumbers/?office=H&election_year=2020)

to the same risk. Although most studies, such as Jacobson's (1978; 2006), focus on expenses made by candidate committees, expenses made by independent organizations are also subject to the same risk. The most profound concern is the question of the presence of reverse causality. Indeed, independent expenditures can sway electoral outcomes, but the opposite holds equally true: electoral outcomes can significantly impact the level of independent spending.

Money may help win votes, but the expectation that a candidate can win votes also brings in money. To the degree that (expected) votes influence spending, ordinary measures will exaggerate the effects of spending on votes. (Jacobson and Carson 2019, p. 67)

Political groups and candidates strategically assess their chances of winning, and their perception of success or failure often influences the amount they are willing to spend. A successful candidate attracts more outside spending, and a well-funded campaign can support a candidate's chances of victory. A clear example of this relationship could be Elissa Slotkin (MI-07) who, despite running in a predominantly Republican district (according to the previously presented PVI), manages to raise a lot of money and win elections (2022). This two-way causal relationship between electoral outcomes and independent expenditures raises questions about the direction of influence, and risks canceling the line between cause and consequence. The relationship between electoral outcomes and independent expenditures might not be as straightforward as it seems. Looking at the study conducted by Gerber (1998) it is possible to notice three different elements that could demonstrate the possible presence of reverse causality:

As the probability of victory rises, it is easier for a candidate to raise money; if the election looks close, supporters are more likely to contribute to the candidates; and if the incumbent becomes sure of victory, the incumbent scales back fundraising activity. (p. 402)

There can be numerous hidden factors that influence the amount of spending decided by independent candidates and committees, like the popularity of candidates, political climate, or the issues dominating public discourse, which could all confound the observed correlation. If these omitted variables remain unaccounted for in the

analysis, it may lead to a false sense of causation. Over the past three decades, researchers have actively explored the relationship between campaign spending and electoral outcomes, utilizing various methods to address potential biases. However, the results have been quite diverse and sometimes contradictory. One consistent observation, though not universally applicable, emerges from research that corrects for endogenous factors such as candidate quality. The common finding, albeit with exceptions, suggests that increased campaign spending tends to translate into greater vote share for challenger candidates but does not have the same impact on incumbent candidates. In other words: challengers who spend more on their campaigns generally see a positive correlation with their electoral success, gaining more votes as a result. On the other hand, incumbents' vote share does not seem to be as significantly influenced by their campaign spending. This finding underscores the importance of campaign finance and its potential to level the playing field for challengers seeking to unseat incumbents. It also indicates that while campaign spending can be a critical factor in determining electoral outcomes, its effects may vary based on the candidates' positions and roles in the race. Further research and analysis are necessary to understand the intricacies of this relationship fully. A nuanced understanding of the context, political dynamics, and timing of events is crucial to identify which variable acts as the driver and which one is the passenger. Yet, it is crucial to recognize that this relationship may not be as straightforward as it appears. The dance between the two variables is marked by complexity, where each influences the other in a subtle, elusive manner. There are analyses, such as the one conducted by Milligan and Rekkas (2007) in reference to the Canadian electoral situation, which try to use the spending limits provided for by the legislation, but which cannot be brought back to the United States, least of all in the case of independent spending given the absence of limits both in fundraising and in their use. Other studies, such as the one done by Erikson and Palfrey (1998), try to overcome this problem by means of a simultaneous equation estimation with zero-covariance restrictions. To untangle this intricate web of relationships, researchers and political analysts face a formidable challenge. Precisely pinpointing the causal direction between electoral outcomes and independent expenditures demands an astute approach that considers the complex interdependencies. For this reason, scholars have tried different methods of approach to the problem through the

use of different methods, but among all the approach with instrumental variables was the one that seemed to give the best results. Instrumental Variables are a powerful statistical technique widely used in econometrics and other fields to address endogeneity problems in causal inference. When studying the relationship between an independent variable and a dependent variable, endogeneity arises when the first one is correlated with the error term in the regression equation, leading to biased and inconsistent estimates. This method provides a clever solution by using a third variable, the instrumental variable, that affects the independent variable but not the dependent variable directly. By leveraging the relationship between the instrumental variable and the potentially endogenous variable, instrumental variables can isolate the exogenous variation in the independent variables, effectively breaking the link between this element and the error term. This way, the instrumental variable produces consistent estimates of the causal effect between the independent variable and the dependent one. However, instrumental variable analysis comes with assumptions and challenges. Proper identification and validation of the instrumental variable are crucial, and the technique might require large sample sizes. Despite these complexities, when employed correctly, instrumental variables analysis can help researchers infer causal relationships in situations where traditional regression methods may fail. The best method to search for an adequate instrumental variable is through the use of pre-existing literature.

Most studies focus on applying these types of models only to the expenses made by candidates during their election campaigns. Given the slight difference in this study, the models previously used will necessarily have to be revised to better adapt to the analysis and the objective that this thesis aims to achieve. The analysis conducted by Gerber (1998) hypothesizes the existence of three possible instrumental variables, all three with an effect on the independent variable but without any clear effect on the dependent one. The three variables hypothesized are the wealth of the candidates, the vote-age population of the state or district where the electoral competition is held, and the lagged expenditures referred to the previous electoral cycle. Of the hypotheses used in that research, the third would seem to be the best, if not the most useful, as it would allow to evaluate the levels of expenditure in the year under review by comparing them with those of the previous electoral round. Not only Gerber, but even the analysis

carried out by Green and Krasno (1988) also uses a form, albeit different, of lagged expenditure to see if, considering the electoral funds previously used, the risk of reciprocal causation that was previously observed could be overcome. As argued by Wang and Bellemare (2020), lagged variables have often been used as instrumental variables because, although they cannot be considered as strictly exogenous, they should be able to alleviate at least in part from endogeneity. These variables have the ability to play a crucial role in various fields, such as economics, finance, and social sciences. Lagged variables refer to variables that are observed or measured at previous time points in a time series analysis. These variables are used to study the relationships between the current state of a system and its past states. By incorporating lagged variables into a model, researchers can capture the dynamic nature of many real-world phenomena, such as economic indicators, stock prices, weather patterns, and more. Lagged variables are valuable because they help researchers analyze the temporal relationships between variables and understand how past values influence present or future outcomes. By introducing time lags, researchers can identify patterns, trends, and potential causal connections that might not be evident with contemporaneous data alone. Moreover, lagged variables are instrumental in addressing issues like autocorrelation, which occurs when data points are correlated with their previous values. By incorporating lagged variables into regression models, analysts can mitigate autocorrelation biases and obtain more accurate and reliable results. Some studies have highlighted the relevance of this type of variable in the context of IVs and many have used spending from the previous election cycle as a tool for spending in the next cycle. Thus, through the use of lagged expenditures performed by independent organizations in the previous cycle as a tool in the approach with IVs, it is possible to overcome the problem of reciprocal causation, as argued by Gerber (1998). But that analysis is necessarily different in that, because the author "*instrumented for campaign spending using lagged spending by incumbents and challengers*" (Sovey and Green 2011, p. 194), while this thesis uses the funds available to the candidates' committees only as a control variable. In reality, however, they are still talking about electoral funds, only the recipient in which they are paid changes. The other notable difference from that study is that this, like Green and Krasno's, refers to election campaigns for the House of Representatives. While the author claims that

Due to the staggered nature of Senate elections³¹, the previous race and the current race rarely involve the same incumbent or challenger. The variable is therefore free from the criticism that might be applied to lagged spending by the same candidate, namely, that specific candidate attributes are correlated with both the regression error and past fundraising levels. Lagged spending should be correlated with the included spending variables for the current election, however, since candidates from the same states may face similar fundraising environments. (Gerber 1998, p. 405)

There is no chance of having the same staggered nature as the elections to the House of Representatives but, looking at the election results of both 2020 and 2018, the year on which the lagged variable employed in this thesis is based, it is possible to observe that while some members remained the same, many have changed, in fact generating new incumbents with each election cycle. Moreover, looking at the report presented by the Congressional Research Service (2022), it is possible to note that the average duration of the mandate of the Representatives is decidedly lower than that of the Senators, showing the presence of a natural turnover in the members that make up that part of the legislative branch. The fact that members of the lower house of Congress are subject to seeking re-election means that they, and the independent organizations that support them, must continually raise funds and spend them again precisely to keep their office. In fact, unlike members of the Senate, they are subject to re-election every two years, while those in the upper house every six give them time to seek funds even during an election cycle in which they do not have to compete with a challenger to keep their seat in Congress. For these reasons, there is no possibility that the funds raised in the previous election cycle can be used in the electoral cycle subject to the analysis. Even the same expenses made by independent organizations are subject to the same factor since, as the FEC data shows, almost all the funds raised were used for electoral campaigns of the candidates that each organization supports.

³¹ The staggered nature of the US Senate election is a key feature of the American political system. Senators serve six-year terms, but their elections are staggered, meaning only about one-third of the Senate is up for election every two years. This setup ensures that the Senate remains a stable institution, with experienced members providing continuity while newly elected senators bring fresh perspectives. Additionally, it prevents sudden shifts in power and allows for a gradual response to changing political landscapes. This system aims to strike a balance between stability and responsiveness, promoting a measured approach to legislative decision-making and representing the diversity of the American people over time.

Moreover, according to the complete list of independent organizations used for the research carried out by this thesis, their number between 2018 and 2020 is decidedly unequal but also the same organizations are completely different. This is justified by the fact that, like the candidates, the organizations themselves also change from year to year, increase in number, and constantly generate new ones. This fact can be simply explained by one factor, which is the simultaneous presence of an electoral cycle that also includes the White House in 2020. So, given the changing situation between the midterm elections and those corresponding to the White House elections, the answer that the system provides is the creation of new independent spending groups or the increase in spending promoted by existing ones. The study aforementioned made by Herrnson study notes that especially after the judgment made by the Supreme Court:

the new state of affairs gave individuals and groups seeking political influence, financial gain, or both a green light to increase their participation in elections. Some responded by creating new political spending groups or augmenting existing PACs or other organizations already active in political campaigns. Freed from previous regulatory constraints, some corporations, labor unions, trade associations, and other groups began to make independent expenditures explicitly advocating the election or defeat of federal candidates.
(Herrnson 2017, p. 7)

The different number of independent groups and the exponential increase in spending made by them is therefore a fundamental part of the analysis. Similarly, the fact that independent spending in the mid-term election cycle is significantly lower than in the cycle under consideration could also indicate other factors. In fact, as the 2018 lagged expenditures are used, both from this thesis and from the works cited, it should serve as a tool to analyze the independent spending of the 2020 elections. In fact, the amount of money corresponds to the prize at stake, which is the control of Congress and of the presidency by a single party. In this, independent committees can be instrumental in allocating the funds they have available in the districts and states most in balance in order to induce the population to vote for a particular party. In addition, as the study conducted by Jackson (2000) notes, the presence of the presidential ticket at the polls induces more people to go to the polls leading to an increase in the number of votes cast. Thanks to the increase in the number of people at

the polls, the independent committees have the sole task of directing votes to the party with which they are most ideologically aligned. For these reasons, using the lagged expenditures made by independent organizations during the previous election cycle could be a good tool to evaluate the expenses made in the cycle subject to the analysis in addition to the fact that having to spend the majority if not all the funds received each election cycle allows to see that no dollar spent in 2018 could have bought a vote in 2020. In addition, using independent spending avoids an additional problem that would arise once campaign expenses are used. Looking at some studies that use campaign expenses made by committees, such as that of Ansolabehere and Gerber (1994) and that of Jacobson (1978), it can be seen that the expenses made by incumbent candidates tend to be less effective if not negative than those of their challengers. The factor that is not taken into consideration, however, if not marginally is that, while the campaigns of the incumbent candidates already have a base from which to start because the electoral machine has allowed them to win at least in the previous cycle, that of the challengers is under construction and in addition they do not have the same recognition of the name of the candidate in office. Consequently, any expense made by the holder of the seat to increase the knowledge of his name and his ideas could be in vain as it would be something already seen in the past. On the other hand, any expense made by the challenger candidate will be functional because he will start with being less known and therefore will have more chances to spread his ideas and his name within the district in which he wants to compete. The incumbents have a political machine already started and do not have to build it, on the other hand, every expense of the challengers will be functional to expand this machine and, therefore, will be more significant. The use of independent expenditures as independent variables and lagged independent expenditures would avoid all the problems mentioned above precisely because, since 2010, independent committees have provided money to the campaigns that are collected during each election cycle. New committees and new money make it possible to analyze the distribution decisions made by these organizations, avoiding the problems that are always present once only the expenses made by the candidates are considered. Another problem that should be avoided is the need to select only a few constituencies, allowing them to be analyzed as a whole. According to Green and Krasno's (1988) research, using the expenses of incumbent

candidates in the previous cycle would force them to eliminate all those cases in which the current members of the House of Representatives were challengers seeking a seat and not already members of the legislative body. But by using the percentage of votes obtained by the incumbent parties, which means considering only the party of the candidate nominated for the challenge of the general election and not the candidate entity in its own right as a member of the House of Representatives as was done in the analysis made by Ansolabehere and Snyder (2003), allows to check for all the expenses made, without having to select only a few cases. Obviously, the expenditure made by the candidates for the elections of November 3, 2020, will be taken into account as a control variable, but the fact that the research considers primarily the role that the expenditure made by independent organizations derives above all from all these factors. At the same time, the use of this type of tool will allow controlling any correlations between the expenditure made in the two electoral cycles as the independent committees that manage to elect the largest number of candidates they support will be more attractive in subsequent cycles, causing donations to them to increase.

Finally, a further element that increases the validity of this instrument is derived directly from a legislative provision. According to the Code of Federal Regulations, it is possible to define independent expenditures

*expenditure by a person for a communication expressly advocating the election or defeat of a clearly identified candidate that is not made in cooperation, consultation, or concert with, or at the request or suggestion of, a candidate, a candidate's authorized committee, or their agents, or a political party committee or its agents.*³²

Therefore, the Commission prohibits any kind of coordination between candidates, or their emissaries, and the organizations that carry out and promote their messages. This factor makes the presence of independent committees even more crucial within the already extremely complex U.S. political landscape as their decisions could favor a certain candidate but also damage him if they were to make certain electoral marketing choices. That is caused by the fact that, if these committees

³² (Federal Election Commission 2014)

were to make choices contrary to those taken by candidates, they could cause distorted messages to reach voters causing numerous problems for electoral campaigns. The judgment issued by the Supreme Court established that political expenditures “*made independently of a candidate or party do not pose a sufficient threat of corruption to justify the infringements on speech and association that result from government regulation of campaign contributions and spending*” (Smith 2012, p. 604). For this reason, the use of lagged independent expenditures referred to the previous electoral cycle allows to control both the increase in spending, but also how this spending affects the election result, as was done in the previously mentioned analysis. The lack of coordination between independent organizations and candidates' election campaigns could constantly influence the choices of both, causing problems if the messages sent are different and, above all if they are conflicting, but also because the contribution of the advertisements broadcast by the two could differently influence the decisions of the electorate, as Wang, Lewis, and Schweidel (2018) also show in their research.

One of the biggest problems associated with this legislation is that, according to many scholars, (Choi 2022; Kang 2012) the legislation of this subject leaves room for many openings, allowing candidates to easily circumvent the constraints. In fact, there are many cases of a certain level of coordination between applicants and independent organizations due to the fact that the legislation, according to the study carried out by Hasen (2014), allows “*the same risks as large contributions directly to candidates, that the Supreme Court’s jurisprudence does not allow*” (p. 2). One of the main methods to overcome these constraints is the use of exploratory committees before the decision to compete for an elected office that allows public figures at all levels to create Super PACs and raise unlimited money simply without having declared themselves candidates. The coordination between independent committees and candidates, therefore, is central to the U.S. public debate but, in reality, the candidates subject to the analysis are mostly immune from these issues for several reasons, including the fact that often the majority in the House of Representatives is not defined by a small number of seats, and not only by a very small number or even by none as happened in the last legislature of the Senate. The fact that every two years the entire lower house is subject to a total electoral renewal means that the electoral map expands considerably, preventing independent commissions from coordinating throughout the

federal territory and forcing them to find targeted targets. Although the number of competitive districts is much smaller than the total, perhaps only candidate-specific Super PACs can direct their funds to a single district, even focusing only on these there are eighty-one districts with a PVI between plus and minus five³³ and therefore considered more competitive, making that, in any case, the electoral map for the lower house is extremely large and expensive. Although the majority of independent organizations are focused on a single candidate, as Dwyre and Braz's (2015) analysis finds, almost all focus on elections to either the Senate or the presidency, leaving it to the most partisan and ideological organizations to carry out the expenses for elections to the House of Representatives. Therefore, the approach that these organizations will use needs to be different so that they can involve as many districts and candidates to allocate as few resources as possible with the purpose of winning as many districts as they can. The work done by Carson and Hitefield (2018) goes in the direction of observing the phenomenon that characterizes the decisions taken by independent organizations in their action. The nationalization of electoral campaigns, especially those for Congress, will be fundamental to achieving the objective set by this thesis.

The nationalization of election campaigns for the House of Representatives allowed congressional candidates to also use messages spread by candidates and/or representatives of their own party in neighboring districts. This fact leads to a logical consequence, whereby when elections are highly nationalized, the probability of Republican legislators representing relatively liberal districts and Democratic legislators representing relatively conservative districts decreases. This indicates that the alignment between legislators' ideological positions and the preferences of their constituents is influenced by the level of nationalization in elections. Notably, legislators representing ideologically mismatched districts tend to adopt more moderate roll-call voting behavior, as Senator Manchin³⁴ and former Maryland Governor Hogan³⁵ tend to do. This suggests that when faced with constituents who hold differing ideological views, legislators are inclined to moderate the ads they will

³³ (The Cook Political Report 2023, <https://www.cookpolitical.com/cook-pvi/2019-partisan-voting-index/116-district-map-and-list>)

³⁴ (Mizelle and Greaf 2023, <https://edition.cnn.com/2023/04/02/politics/joe-manchin-2024-plans-cnn/index.html>)

³⁵ (Kilgore 2022, <https://nymag.com/intelligencer/2022/01/larry-hogan-and-the-fantasy-world-of-moderate-republicanism.html>)

use for their electoral campaign in order to strike a balance and represent the diverse interests of their district. The phenomenon of ideological moderation in such cases underscores the dynamic nature of representation and the efforts of legislators to navigate the ideological diversity within their constituencies, ultimately seeking to effectively represent the interests and preferences of their constituents. On the contrary, what is currently happening in the US system is that most Republican districts vote only for Republican candidates, while most Democratic ones vote only for the representatives of that party. This factor is crucial in the political landscape as it causes candidates not to present their ideas but to focus on those of the presidential candidate since, as Hoffman's (2013) study notes,

voters are focusing on the President's policies and other issues that affect not only their district or state, but also the entire country. The President and his national policies have become a major point of concern to voters when they decide whom to vote for as their Senators and Congressmen. Legislative elections are influenced by the national-level trends running in favor of one majority party or Presidential candidate. (pp. 3-4)

Thus, all candidates for Congress refer to party policies and no longer to their ideas for the district or the state that remain valid only to a small extent. In addition, this argument can also be observed in the election advertisements that independent committees and organizations use. In fact, the central themes become the support of one or the other candidate, or their support for a certain policy that the presidential candidate intends to promote. Often it is also the presidential candidates themselves who use these types of messages to promote a particular candidate that they consider most favorable, using a type of language that equates a vote for a candidate for Congress with a vote for the President or candidate for President. “*A vote for Marsha is really a vote for me and everything we stand for. It's a vote for 'make America great again.'*”³⁶ said then-President Trump speaking of Senate candidate for Tennessee Marsha Blackburn. The nationalization of electoral campaigns, therefore, brings with it the abandonment of local issues preferring to discuss more general topics that

³⁶ (Ye Hee Lee and Sullivan 2018, https://www.washingtonpost.com/politics/in-deep-red-tennessee-republicans-are-anxious-about-the-us-senate-race/2018/10/04/de13d5a6-c57d-11e8-b2b5-79270f9cce17_story.html)

concern the entire federal territory. This decision, made during Reagan's two presidential terms, meant that candidates at all levels relied on the President instead of their constituency. As Sievert and McKee's (2019) analysis notes, the nationalization of the countryside has had important consequences. Firstly, it has led to a reduction in ticket-splitting as with the same programs people will tend to vote for the same party and, secondly, it has decreased the importance of the personal factors of local candidates while increasing that of the presidential approval rating. Many scholars have tried to hypothesize the causes of this alignment between voters and parties and have often hypothesized that the problem lies not in citizens aligned with the parties but in the fact that *"each party's candidates now look alike, so voters have much less reason to split their tickets. Few voters have a liberal Republican or a conservative Democrat to vote for today"* (Fiorina 2017, p. 1). However, not only the alignment of candidates with parties has helped but also the commonality of ideas, opinions, and above all electoral messages transmitted. In an era in which U.S. citizens are filled with campaign messages every day not only during the campaign period, the pooling of campaign messages and advertisements has allowed messages to expand faster while also covering a much larger portion of territory.

The entirety of the United States is divided into two hundred and ten television market areas which, therefore, do not correspond to electoral districts. The U.S. television market is divided across congressional districts, as noted in the analysis made by Stewart III and Reynolds (1990), reflecting diverse viewer preferences. Urban areas tend to have concentrated media markets, while rural districts may have limited options. This division shapes media outreach during elections and influences programming to cater to local demographics, creating a mosaic of broadcasting landscapes. The fact that many districts share the same advertising marketing linked to the increase in nationalization of electoral campaigns, means that candidates and independent committees use the same messages, or at least similar, in order to reach the widest possible audience. As can be seen in the study of Moskowitz (2018), the geographic distribution of media markets creates a situation where certain voters are exposed to a neighboring state's media market, while others are within their own state's media market. This variation in media exposure can have an impact on the information voters receive and the political narratives to which they are subjected. Voters residing

in neighboring district markets may be more influenced by news and advertisements related to political races and issues in that neighboring district, potentially shaping their perceptions and political attitudes differently compared to voters within their own media market. This geographic distinction in media markets highlights the complexity of voter information and the potential for varying levels of exposure to different political messages based on geographical proximity. Understanding this geographical influence on media exposure is crucial in comprehending the nuances of voter behavior and the formation of political opinions across different regions. In fact, as described in the analysis made by Nagel (2001), in recent years, there has been a crucial phenomenon for U.S. policy: the election campaigns, especially those for Congress, began to convey information and links no longer only with the electoral district of the challenge, but with the political environment in general, uniting the political messages of all districts. This fact, caused mainly by the use of images of presidential candidates or leaders of the two parties in Washington, meant that the messages were united and that the candidates of a certain district transmitted messages mostly similar and with a broader scope. Many studies have focused on assessing the impact of ads on election outcomes (Huber and Arceneaux 2007; Sides, Vavreck, and Warshaw 2022) noting that exposure to election marketing has also helped increase the number of voters but have noted that this sharing has amplified the phenomenon of campaign nationalization. Other studies, such as the one done by Holliday (2022), have noted that in non-debate contexts (therefore both in advertising and on social networks) the themes and tones of local electoral campaigns are superimposable to those of presidential campaigns or in any case to those of the incumbent President. For these reasons it is possible to hypothesize that each electoral campaign plays a crucial role in the neighboring ones both thanks to the diffusion of similar electoral messages referring to the same personalities, and for the use of the same channels to broadcast the advertisements that the candidates, and especially the independent organizations, buy on the television networks. Independent spending, therefore, goes in the direction of spreading homogeneous electoral messages to exploit both these factors in order to reach, as mentioned above, the largest number of voting citizens and direct them to vote for the candidate they support. In addition, the purchase of advertising space on social network platforms could also help the spread of this type of message. All this,

combined with the fact that districts tend to be increasingly clustered geographically with Democrats getting the most votes in the east and in large cities while Republicans, on the contrary, get them in the west and in more rural areas, as noted by Kim, Elliott, and Wang (2002), may have influenced the spending decisions made by the independent committees, pushing them towards some decision of how and where to allocate their fund rather than opt for another place or candidate. For these reasons, the evaluation of the distance between districts is fundamental to achieving the objectives set by this work. As the research made by Bodet, Bouchard, Thomas, and Tessier (2022) notes “*the electoral district is the fundamental unit of representation in single- and multi-member electoral systems, yet most research shows little interest in district effects on election outcomes, focusing instead on national and individual factors*” (p. 154). And the fact that the units of analysis of this thesis are the districts of the House of Representatives helps to better evaluate the factors previously presented through the application of a spatial regression that takes into account all the elements already mentioned earlier in the course of this chapter, as was done in the research made by Crespin, Darmofal, and Eaves (2011) that stated that the “*Congressional elections are not independent contests*” (p. 1).

Over the years, many researchers have decided to try their hand at applying a spatial model to evaluate election results, some with the aim of observing the expenses made by both committees and organizations independent of them. This is the case of a study already mentioned above, the one done by Ferguson, Jorgensen, and Chen (T. Ferguson, Jorgensen, and Chen 2019), which uses a spatial approach to assess the effect of spending on election results. As mentioned, the proximity between the districts and the nationalization of electoral campaigns has allowed a general standardization of the messages spread by all the actors involved and this has meant that the ideas of one or the other organization propagated even faster. The growing interconnectedness between people, aided mainly by the enhancement of the means of communication and the development of infrastructure, has helped election campaigns enormously. The gradual increase in studies that apply models that also take into account the spatial distance between districts, always with the aim of taking into account this factor to calculate which are the ones that most influence the outcome of the elections, is fundamental for the progress of the work of this thesis. In fact, by

applying this variable to the expenses made by organizations outside the candidates' committees in disseminating electoral messages with a negative value, it will be possible to estimate the role played by independent expenses during the elections of 3 November 2020. Spatial analysis plays a pivotal role in understanding the impact of independent expenditures on the 2020 House of Representatives elections. By incorporating spatial analysis techniques into the thesis, researchers can unveil critical insights that would otherwise remain hidden in traditional data analysis. The 2020 House elections witnessed an unprecedented surge in independent expenditures, making it crucial to employ spatial analysis to investigate their influence on electoral outcomes. As was said in the research made by Su and Qu (2010), spatial analysis allows researchers to discern patterns, trends, and correlations that are geographically distributed. By mapping the locations of independent expenditure funding and analyzing the demographic characteristics of the affected areas, researchers can identify potential hotspots of influence, voter behavior, and partisan shifts. Furthermore, spatial analysis facilitates the identification of gerrymandering, a pressing issue in U.S. elections³⁷. It helps expose manipulative redistricting strategies, which can significantly impact electoral fairness and representation. Understanding the spatial dynamics of independent expenditures aids in identifying marginalized communities that might be disproportionately affected by outside influence, offering valuable insights into the state of democracy and the potential for political inequalities. Utilizing spatial analysis in a thesis about the role of independent expenditures in the 2020 House elections empowers researchers to reveal nuances and connections that standard analyses might overlook. By illuminating spatial patterns, this analytical approach enhances our comprehension of the underlying factors influencing election outcomes, ultimately contributing to a more informed and holistic understanding of the electoral process, as Gibson (2018) tries to do in her thesis. Despite the nationalization of campaign messages for seats in the lower house of Congress,

³⁷ Gerrymandering, a contentious political practice, involves manipulating electoral district boundaries to gain an advantage for a particular party or group. By strategically drawing these boundaries, politicians can dilute or concentrate voters of opposing views, distorting the democratic process. This undermines fair representation and can perpetuate the power of incumbents, stifling political competition and public accountability. Gerrymandering often leads to skewed election outcomes, disenfranchising citizens and eroding trust in the democratic system. Addressing this issue requires nonpartisan efforts to establish impartial redistricting processes, ensuring that the voice of every voter is equally heard and respected in shaping the future of their communities and nation.

representatives must necessarily maintain a link with their district and the use of geospatial techniques allows to analyze how much this link explains the election result. The evolution of computing machines has also allowed the development of very rapid geospatial models and techniques that allow to analyze much faster and easier the endogenous problems deriving from the presence of these types of factors, applying this type of measurement also to elections, as argued by Byers and Gill (2022) in their work in support of the application of geospatial models in the era of Big Data. As was previously mentioned, the geospatial analysis conducted by this thesis will be based on the districts of the House of Representatives, going to consider all the variables previously observed in this chapter and more fully described within the next. The districts will act as a basis for the analysis as all the features listed above will be observed using them as a starting point. Therefore, the application of a geospatial technique would allow to evaluate the possibility that these results go, at least indirectly, to influence the outcome of the 2020 elections. The application of a model of this type is again crucial, especially if crossed with the other values observed, as it would allow to better evaluate its effect on the final result of the elections. As noted by the review of studies using an instrumental variable by Betz, Cook, and Hollenbach (2020), there are only a few who try to use a spatial approach, although this type of model has largely proliferated within the study of political science. Ignoring the geospatial interdependence in the analysis of electoral outcomes can lead to distorted conclusions and flawed policymaking. In elections, the spatial context matters as neighboring regions often exhibit similar voting patterns due to shared demographics, cultural influences, and economic ties. Failing to account for these interdependencies may overlook crucial factors influencing voting behavior and result in inaccurate predictions. Spatial autocorrelation and regional spillover effects are often overlooked, leading to a limited understanding of the broader impact of electoral results. Embracing geospatial analysis techniques allows for a more comprehensive and nuanced assessment, leading to better-informed decisions in the realm of politics and governance. Another research conducted by Franzese and Hays (2007) observed that neglecting the significance of spatial interdependence as a mere nuisance poses challenges both from an econometric perspective and, more crucially, from a substantive standpoint, as numerous contexts in political science involve substantial

and intriguing interdependencies among units of observation. These models account for spatial interdependencies, capturing the indirect effects and contextual factors that influence the relationship between variables. One key advantage of using a spatial regression model is its ability to capture the spatial autocorrelation present in electoral data. While, traditional regression models assume that observations are independent, which can lead to biased estimates when dealing with spatially correlated data, spatial regression models, on the other hand, incorporate spatial weights matrices, both dichotomous indicating whether one district borders another but also geographical distances between constituencies, that account for the influence of neighboring units, providing more accurate and robust parameter estimates, as explained in the work of Pace and Barry (1997). Moreover, spatial regression models allow researchers to examine the spatial spillover effects of independent expenditures on electoral outcomes. By considering the potential diffusion of campaign spending across nearby districts, these models offer a more comprehensive understanding of how independent expenditures impact election results beyond the immediate target district. Additionally, spatial regression models can handle various data types, including continuous, categorical, and spatially explicit variables. This versatility enables researchers to incorporate a wide range of factors influencing electoral outcomes, such as demographic characteristics, economic indicators, and past voting patterns, into a unified analytical framework. Employing spatial regression models to assess the role of independent expenditures in electoral outcomes brings significant advantages in terms of accounting for spatial dependencies, capturing spillover effects, and accommodating diverse data types. By leveraging these strengths, researchers can gain insights into the complex dynamics of electoral processes and make more informed decisions for campaign strategies and resource allocation. As explained by Cutts and Webber (2010)

while socio-economic factors and local campaigning are key determinants of party vote shares in constituencies, there is strong spatial autocorrelation in voting patterns. Not only does local campaigning have a positive effect on party performance in constituency j , but also empirical evidence is found of spillover effects following intensive party activity. The more a party spends on

campaigning in constituencies adjacent to constituency j, the more votes it gets in constituency j. (p. 183)

Therefore, starting from the analysis unit also used for the other models, this work will evaluate the impact that independent spending had on the elections to the House of Representatives in 2020. The data set used to develop these models, as will be explained in more detail in the next chapter, will contain four hundred and thirty-six observations for each of the factors previously described that is, one for each of the seats in the lower chamber of Congress plus that relating to the non-voting delegate of the District of Columbia. The same election consultants, the same independent committees, the same electoral strategies, and the same political environment, seasoned with the presence of votes for the President of the United States, may have played a crucial role in determining the outcome of the November 3, 2020 election. Therefore, the application of a spatial regression model could help even more to understand the results of that vote also by evaluating when the proximity or distance between the districts has increased the electoral result of one or the other party. Thanks to the rapid development and presence of geographic information systems (GIS), as noted by the analysis of Cho and Gimpel (2012), the knowledge and practical applications of spatial measurements have greatly increased and will be used in this thesis precisely with the aim of observing the close connections between the U.S. districts. Unlike many analyses that focus on the outcome of the presidential election to apply a spatial regression, in this one the result of one of the houses of Congress will be analyzed as done by Bowen (2014), also taking into account the main element that links U.S. geography and politics, the redrawing of electoral districts according to the principles of gerrymandering using the PVI.

The work that will be developed is inspired by all the literature so far presented, differentiating itself in some quite relevant aspects with the purpose of innovating some elements, adding some that had not been previously considered, and eliminating those considered superfluous both for the changed political conditions and for the increase in the presence of funds in US electoral campaigns. The first innovative element to take into account is the fact that the analysis will be conducted using data referring to a single year, without looking at the historical progression or the role of all previous elections. Considering only one election and focusing only on the districts of

the House of Representatives avoids the need to have redundant elements, such as duplicative campaign efforts and excessive allocation of resources to cover multiple elections concurrently (Rocca and Clay 2021; Scala 2020) or not to consider other possible elements that could affect the outcome of the election, such as varying state-level policies, national political trends, or unrelated external factors (Abramowitz and Segal 1992; Levitt 1994). By focusing solely on the districts of the House of Representatives in a single election, analysts can more accurately attribute shifts in voter sentiment and outcomes to the specific dynamics within those districts. This approach reduces the complexity introduced by extraneous variables and provides a clearer understanding of how constituents' preferences align with the candidates' platforms at the local level. It also facilitates a more straightforward evaluation of the effectiveness of campaign strategies and allows for a more precise assessment of the representatives' performance based on the unique characteristics of their constituencies. The use of only one election year, in addition, allows us to better understand the role played by the expenditure made by independent organizations at that juncture. Although there are many studies cited so far that deal with the role of spending in electoral campaigns, only a few of them deal with elections to the House of Representatives both because in each electoral cycle, all its members must be re-elected, and because this continuous innovation leads it to be subject to many other external factors, such as the approval of the President, the presence of other elections, but also the economic situation and the assessment of the policies implemented. Moreover, most of the analyses observed focus on the role of candidates' expenses, i.e. those of electoral committees that candidates manage directly, without focusing on the role of independent spending. These, although with the differences mentioned above, can be combined with the expenses made by the committees and analyzed separately to visualize their respective role in electoral campaigns. Since the Supreme Court ruling, as mentioned, these have grown exponentially and have played an increasingly fundamental role in the electoral process from the primaries to the general elections (Boatright, Malbin, and Glavin 2016; Boatright and Zachary 2021). This thesis, again, aims to investigate the role played by the independent organizations during the 2020 election cycle, with the aim of understanding whether or not the expenses made by these committees helped the election of candidates in their respective districts.

Starting from all the literature cited so far and through the analysis carried out in the following chapters, the objective that this thesis aims to achieve is to try to innovate the research done so far by inserting some new elements starting from the variables presented so far. There are some factors that can be considered innovative and all of them will be used to achieve the purpose of this work. The first innovative element is how this work will consider the incumbency status: while the large majority of the previously mentioned analysis uses the incumbency status of the candidates, dividing the candidates in each district between incumbent, challenger, and candidates in open districts, this work will use the incumbency status of the parties through the use of the outcome of the 2018 elections to the House of Representatives to observe if the results of the previous electoral cycle increase the chance of victory of candidates in the general election, or if almost there could be reverse causality between the dependent and the independent variable of the analysis. Therefore, using the result of the 2018 as a control for the outcome of the 2020 election would allow to observe if there is a possibility of having reverse causality supported by the expectation of winning caused by the outcome of the previous cycle. Moreover, as has been widely seen, there is a substantial difference between the challengers and the incumbent candidates, both in the way of campaigning and in that of approaching independent committees. For these reasons, the use of the two-party vote share referred to the 2018 elections could be useful to observe voting trends, especially in a political system as polarized as the United States. In fact, as noted by Lachat (2008) “*polarization reinforces the impact of ideology on the vote*” (p. 17) proving that the incumbency status, also of the party, could influence the outcome of the election.

Moreover, another element widely used by political analysts and local legislators is the Partisan Voting Index. The use of this index provided by The Cook would allow to understand if the historical voting pattern has any type of influence on the actual electoral result. By using this index, the purpose is to observe if the partisanship of each electoral seat could help or damage the candidacy of a certain party in the districts, with the idea that the more the index is high or low, the less the probability for the other party to win that seats. Therefore, this index would evaluate the polarization of the political landscape, because if a party is not able to win a district that tends a bit towards the other party, it will be difficult for them to win the majority

in the lower chamber of Congress. Precisely thanks to the fact that the PVI measures how much a certain district has voted more for the Republican party or for the Democratic one during the past two presidential elections, it will be possible to understand if the partisan leaning of the districts will affect the election outcome by helping one or the other candidate. In addition, also considering the result of the presidential candidates should help to better understand the result of the elections in this sense, while allowing to assess if and how much the ticket-splitting is present and be able to determine the composition of the House of Representatives.

The last innovation is probably the most important and is also one of the main elements of this thesis. Although there are many studies that focus on the use and role of spending within electoral campaigns, there are very few that make a distinction between the candidates' funds and the external help provided by independent organizations despite the fact that after *Citizen United* versus FEC the literature in this sense has increased a lot. This thesis will take into account both the role played by the funds available to the candidates' electoral committees and those who, at least at the legislative level, should not coordinate with the candidates. The expenses made by candidates and independent candidates will be divided into two categories each, that is, depending on whether they were made in favor of a Democratic candidate or a Republican candidate, following what is established with the incumbency status. In contrast, many other papers evaluate spending as a differential between what Democrats do and what Republicans do or vice versa. Considering them as two distinct elements could help to better understand the role they played. Starting from the work of Engstrom and Kenny (2002), who already noted the important role of independent expenditures in the elections between 1984 and 1996 in influencing voter behavior and choices, there will be an attempt to try to interpret the role played by these expenses during the elections of 3 November 2020 that were somewhat historic. Consequently, using divided expenses allows both to evaluate the effect individually on a single variable dependent on the differential between the votes obtained by the Republicans, who will be used as a basis, and the Democrats. The dependent variable of this thesis, in fact, will follow what has been done in the work of Ferguson, Jorgensen, and Chen (2019) using a differential that will have positive values in case they have won Republican Party candidates and negative in the case of Democrats.

Chapter 2 – Data and Method

There's money on both sides of every issue in America. – Marco Rubio³⁸

As previously explained, the entire second chapter of this thesis is dedicated to the presentation of the data and methods used to conduct the analysis. The chapter will then be divided into two sections: in the first one all the variables used will be described, while in the second the three models used by the research will be presented. All the data on which the analysis is based comes from U.S. government websites, mainly the archive of the FEC which collects funds from both candidates and independent organizations for more than forty years, or from online archives that collect all election results. Since the theme of the study refers only to 2020, all the data used will be cross-sectional and secondary. Cross-sectional data analysis is a research approach where data is gathered from a diverse sample of individuals, groups, or entities at a particular moment, and, unlike longitudinal studies, cross-sectional data doesn't track changes over time, but it offers a snapshot of a certain moment, that in this case are the 2020 elections. Researchers use this method to examine relationships between variables, explore trends, and draw conclusions about the population at a specific point in time. This type of data is valuable for quick assessments, identifying patterns, and generating hypotheses. However, it cannot capture temporal changes, and it's crucial to be cautious about establishing causal relationships based solely on cross-sectional data. These data will be used in the models that the second section of this chapter will explain in detail. The three models implemented in this thesis are the OLS (Ordinary Least Squares) Regression, the IV (Instrumental Variable) Regression, and the Spatial Regression, which are distinct but important techniques used in econometrics and statistics to model and analyze relationships between variables. The OLS Regression is a widely used linear regression technique that aims to estimate the relationship between a dependent and one or more independent variables. IV regression helps to establish causality when the relationship between the dependent and independent variables is potentially confounded by endogeneity. Spatial regression models consider the spatial structure of many real-world scenarios and help incorporate the spatial context into the analysis.

³⁸ (CNN Town Hall on Gun Violence 2018, <https://edition.cnn.com/2018/02/21/politics/cnn-town-hall-florida-shooting/index.html>)

Data description

As mentioned in the introduction of the second chapter of the thesis, the whole structure of this section is based on the presentation and description of the data employed in the models used to perform the analysis of the role that independent expenditures have had in the outcome of the elections to the House of Representatives on November 3, 2023. Each of the data used is secondary data which, according to Clark (2005), is *“information that has already been collected by someone else and which is available for [...] the researcher, to use.”* (p. 57). There will be eight sections for the data descriptions that will take into account all the elements presented in the introduction, they will be: the independent expenditures, the expenditures made by each candidate, the Partisan Voting Index provided by the Cook, the geographical boundaries of each of the four hundred and thirty-five districts where the Representatives reside plus the District of Columbia, the outcome of the 2020 House of Representatives elections, the outcome of the 2020 Presidential elections, the outcome of the 2018 House of Representatives elections, and the independent expenditures made during the 2018 midterm elections. Through a careful reading of this part of the chapter, it will be possible to understand both how the variables will be structured and the motivations that led to their use. In addition, it will be necessary to take into account that each of the final values used has followed a machining process, which means that through the use of R, each variable has been created starting from a previous data set provided by an organization that will be mentioned in each section. The use of summaries and graphs through the whole structure of these paragraphs will help to better understand the distribution of variables used to achieve the purpose of this thesis.

Independent expenditures – 2020

The expenditures made by independent organizations are all collected within the website of the Federal Election Commission³⁹, an independent regulatory commission composed of seven members of presidential nomination and Senate confirmation of which no more than three can come from the same political party. The Commission is responsible for tracking the movement of money to and from all

³⁹ (Federal Election Commission 2020c, https://www.fec.gov/files/bulk-downloads/2020/independent_expenditure_2020.csv)

organizations that use it for political purposes and makes available on its website all the bulk data relating to any kind of money received or spent by both candidates' election campaigns and independent organizations. The data set provided by the FEC contains all the independent expenses made during the 2020 electoral cycle. Once all the expenses not related to the elections under consideration in the current analysis are removed, the data set obtained contains 17,579 observations related just to the general elections for the House of Representatives. Since the unit of analysis of the thesis is the expenditures made in each district in favor of both parties, these observations will be added taking into account these two factors so as to obtain two variables, one per party, for each constituency. In addition, it is important to specify that within the data set there was a qualitative variable that defines the purpose for which the independent organization examined went to make that expense. After observing each value, it was decided not to eliminate any of them because any expenditure made by the independent committees is functional to their attempt to elect or hinder the election of a certain candidate. After completing the creation of the two variables useful for the analysis, the resulting data set contains four hundred and thirty-six rows, one for each district plus the District of Columbia, and four variables, two for the expenses made by independent organizations to support candidates in the general election and two containing the names of the two challengers of that district. The lack of independent spending made for or against candidates of a party other than Democratic or Republican made it possible to consider just the independent expenditures made in order to support or oppose candidates of the two main parties in the general election. The greatest criticality highlighted in this passage was the presence of some districts of California, Washington, and Louisiana where two candidates of the same party clashed in the general elections, because of the different electoral systems that have these states. Thanks to the work done by Hirano and Snyder (2019), the problem of having internal challenges within the same party has been solved by considering the incumbent as a member of his party and the challenger as a member of the other in order to have the opportunity to evaluate the contribution that being in office brings to the candidates. As explained in the introduction and in the previous chapter, the independent expenditure made for both Democrats and Republicans will be used as the main independent variable. The distribution of expenses on the U.S. territory is

quite different among all the districts: the one made in favor of the Democrats varies between zero and about twelve and a half million, while that in support of the Republicans varies between zero and twelve million, in both cases it refers to the second district of New Mexico. The most competitive districts are those most affected by spending, as they are also those that will then define who will be in the majority. If all districts are taken into account, the average independent expenditure for Democrats is less than eight hundred thousand dollars per district, while that of Republicans exceeds eight hundred thousand. This could be justified by the fact that the Democrats had a majority and the election forecasts claimed that they would have a gain of seats allowing them to concentrate their expenses only in some districts. Instead, Republicans had to defend their seats hoping to win some to weaken the Democratic majority. As Dwyre and Braz's analysis notes (2014), independent organizations *“spend their money differently than conventional PACs in that they are less interested in access to sitting lawmakers and more focused on an electoral strategy to affect the partisan composition of government.”* (p. 684). Therefore, both of them will be considered the two main independent variables of this thesis.

Candidates' campaign expenditures – 2020

As for independent expenditures, the FEC makes bulk data related to expenditures made by candidate campaigns available on its website⁴⁰. In this case and unlike the independent expenses that must indicate who makes them and the candidate who would benefit or harm them, the passage is direct as it is the committees of the candidates who receive and spend the funds. Therefore, the file downloaded from the website of the FEC contains one record for each campaign for the House of Representatives and for the United States Senate and each record contains a summary of the financial information regarding each candidate to the Congress. Differently from the previous dataframe, it is necessary to make some more important decisions regarding which observations to keep and which, instead, can be eliminated from the dataframe. Taking into account the independent expenditures and the election results that will be analyzed in the following pages, the decision taken was to consider only the funds spent by the two candidates who took the most votes in the elections, therefore basically the candidate of the Republican Party and that of the Democratic

⁴⁰ (Federal Election Commission 2020b, <https://www.fec.gov/files/bulk-downloads/2020/web120.zip>)

Party, except the previously mentioned case in California, Washington, and Louisiana, in which two candidates of the same party compete. In an extremely limited number of cases, the incumbent candidate has not received any opposition and, therefore, in twenty-one cases there will be only one candidate in the district, with just one expenditure provided by his or her committee. Looking at the objective of this thesis, the expenses made by the candidates' committees could be decisive for understanding the outcome of the elections and, therefore, will be used as a control variable within the analysis. After eliminating all the remarks referring to the campaigns for the Senate, those of the primary elections, and those referring to the candidates of third parties, in order to keep just the two main candidates, it was decided to keep only three fundamental variables that will be used to observe the influence of those factors on the electoral outcome. These are the districts in which the campaign took place, and the total expenses made by the electoral committees of both the Democratic candidate and the Republican one. According to the FEC, for the general elections to the House of Representatives on November 3, 2020, more than one and a half billion dollars were spent⁴¹ by the eight hundred and fifty-one candidates for the lower chamber of the US Congress, with an average of almost two million dollars spent by each of them, that is more than three and a half million spent on average in each electoral district. The summary of the expenditure indicates that the minimum expenditure made by the Democrats was zero, while the maximum was more than twenty-two million dollars, spent by the then Speaker of the House Nancy Pelosi (CA-12). On the other hand, the Republicans spent in each district between zero and more than thirty-three million dollars, spent by then Minority Whip Steve Scalise (LA-01). As for the independent expenditures, there should be a relationship between the increasing amount of spending and the district's competitiveness index (Dwyre and Braz 2015). However, the fact that the two districts with the highest expenditure are two solidly partisan raises some doubts about this relationship. Actually, party leaders are always the first in terms of funds raised and consequently are also the first in terms of spending made both in their district, which is almost always solidly in the hands of their party, and in the most competitive districts that the parties want to conquer or must defend. The biggest

⁴¹ (Federal Election Commission 2020a, https://www.fec.gov/data/spending-bythenumbers/?office=H&election_year=2020)

problem that arises in relation to the use of this data is related to the place where the expenditure was made. Since the FEC does not indicate where the candidates' committees make the expenditure, it relates only to the district in which the candidate who spent the money is located. As can be seen through those few lines, the expenses of the candidates for the House of Representatives will be divided between those made by the Republican candidates and those made by the Democratic, and, unlike other papers that consider spending between the incumbent candidate and the challenger (Jacobson 2015; Erikson and Palfrey 1998), this analysis will consider the candidate's party to have done the shopping, as Gaddie's (1995) did. At the same time, the incumbency status will be taken into account both through the analysis of the previous election results, and as a dichotomous control variable that will indicate the party of the incumbent candidate in the district.

Cook Partisan Voting Index

The Cook Partisan Voting Index⁴², also known as PVI, is a measurement of how strongly a district or a state leans toward one of the two major parties. PVI index values are determined by contrasting the candidate's average share of the two-party presidential vote in the preceding two presidential elections with the party's national average two-party share for those respective elections. The index is constructed and modified after each presidential election, but also when a state redraws the boundaries of the districts of the House of Representatives. The values within it range from more Democratic to more Republican and are written with a letter (D or R) and a positive value indicative of how Republican or Democratic that district is. All districts that fall between 0 and 0.5 are defined as "even" and are to be considered the most contested districts in the elections. In addition, it is the same portal of political analysts to add that the most swing districts are those with a PVI value between 0 and 5, whether they are more for Democrats or Republicans. For the purposes of this thesis, the decision taken was to use the index as a continuous value from negative to positive depending on whether it is more inclined towards the Democratic or Republican party. The distribution of the index, after making it continuous, varies between minus forty-four, the most Democratic district is the fifteenth in New York, and more thirty-three, the

⁴² (The Cook Political Report 2023, <https://www.cookpolitical.com/cook-pvi/2019-partisan-voting-index/116-district-map-and-list>)

district that tends most towards the Republican Party is the thirteenth in Texas. The values of the Cook Partisan Voting Index are universally used by politicians and analysts especially when there is a need to redraw electoral districts, but also when they have to decide how to allocate funds for elections and organize events related to campaigns. Similar to what was done in the analysis of Nielson and Visalvanich (2017), PVI will be used in models to control district partisanship. In addition, as shown in Figure 1, with the division of the index into seven distinct categories (safe, likely, and lean Dem or Rep, and tossup⁴³) it is possible to visualize the amount of spending made by both independent committees and those of the candidates in the November 3, 2020 elections.

⁴³ (Sabato and Kondik 2013, <https://centerforpolitics.org/crystalball/articles/ratings-changes-and-non-changes/>)

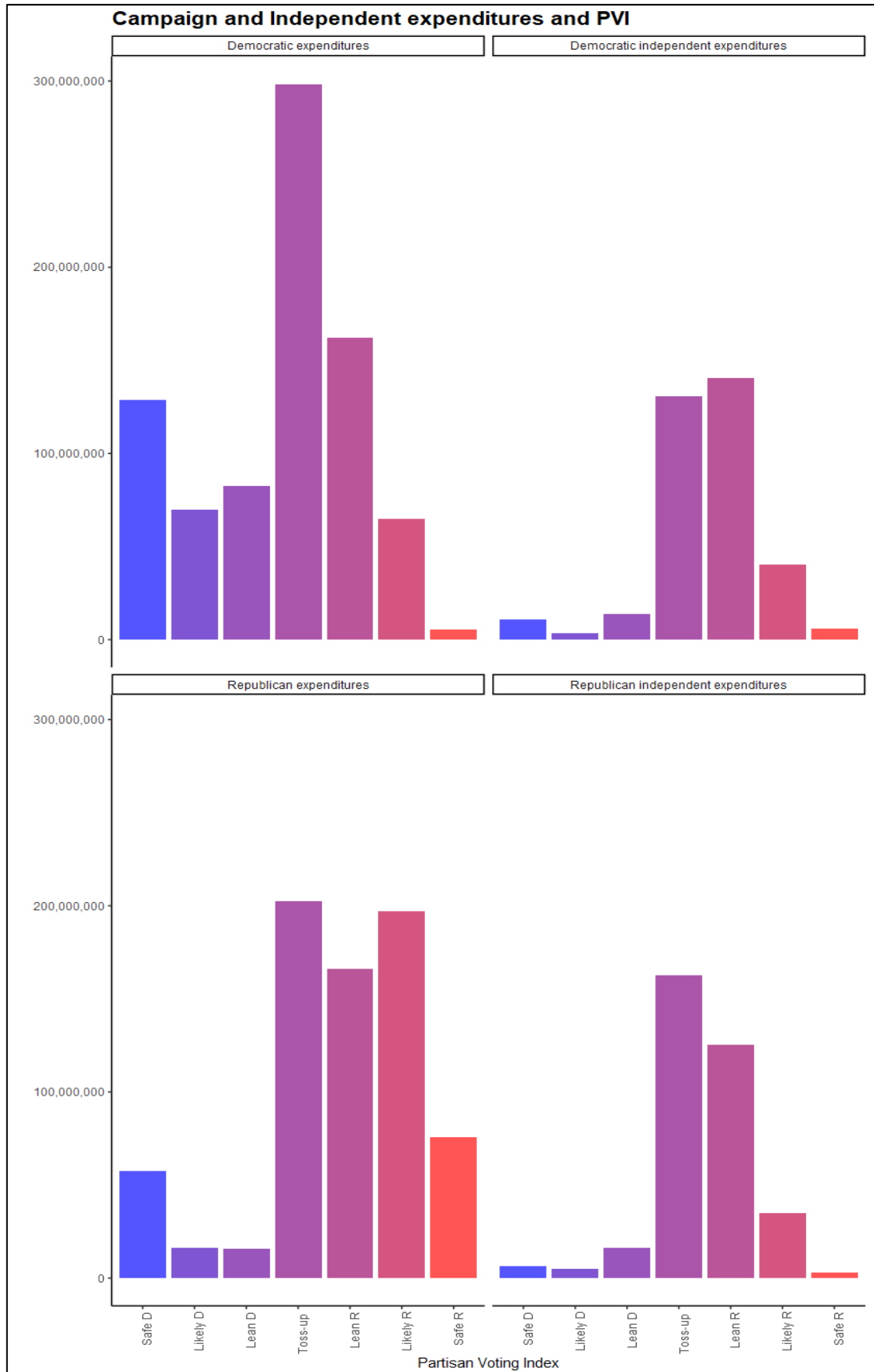


Figure 1 - Expenditures distribution according to the PVI

Congressional districts of the House of Representatives

The shape file for the 117th Congress was downloaded from the ArcGIS site⁴⁴, a geographic information system (GIS) produced by Esri. The dataframe constructed through R's sf package is provided with a column containing the geographic boundary data of each of the four hundred and thirty-five U.S. districts, plus territories that are under U.S. control such as Puerto Rico, Guam, and the District of Columbia. All territories with a non-voting delegate have been eliminated, except for the District of Columbia as this can be useful for research, both because of its proximity to other districts and because of the presence of presidential elections there as well. Starting from that information, two different matrices have been constructed, one containing the geographical distance between the districts and a dichotomous one with 0 and 1 indicating whether or not the district borders another. Both matrices will be tested to decide which is better for the development of the model presented in the next chapter. In addition, starting from the work of Neumayer and Plümper (2016), the result of both the matrix will be considered with and without row standardization in order to use the model that will turn out to be better. In the second section of this chapter, all the methods used to carry out the analysis will be set out and adequately justified.

The results of the 2020 election – House of Representatives

As was recounted in the introduction of this thesis, the results of the elections to the House of Representatives were quite surprising: while the Republicans simultaneously lost the White House and awaited the result of the runoff in Georgia, they managed to reduce the disadvantage in the number of seats they had in the lower house of Congress (Johnson 2021). The election result shows that the Democrats have fallen from the previous two hundred and thirty-five seats won during the 2018 midterm elections to two hundred and twenty-two seats plus the non-voting delegate of the District of Columbia in the 2020 election. At the same time, the Republicans rose from one hundred and ninety-seven to two hundred and thirteen. The results confirmed the majority of the House of Representatives in the hands of the party of then-Speaker Pelosi, but greatly reduced their advantage. In fact, these results were somewhat historic as it was the first time since 2004 that Republicans gained seats in the House of Representatives and handed the Speaker the narrowest Democratic

⁴⁴ (esri_dm 2021, <https://www.arcgis.com/home/item.html?id=f3bd93c389d64813aa7df2f88f51967a>)

majority since 1942. The result of the elections to the House of Representatives was taken from the website of the Massachusetts Institute of Technology which in the Election Data Science Lab collects the election results of both the houses of Congress and the presidential elections since 1976⁴⁵. The dataframe used contains the votes of all the candidates to the House, also those that represent third and minor parties, and the decision taken in this case has followed the one taken previously. In view of the fact that independent expenditures were made only in relation to the candidates of the two main parties, the choice was to keep only those two parties within the analysis that will be conducted. After reorganizing the dataframe, the obtained table presents four variables that are the electoral district of the candidates, the votes obtained by the Democratic candidate, the votes obtained by the Republican candidate, the sum of the votes obtained by those two, and a column with a dichotomous variable indicating the party of the winning candidate. For the purpose that this thesis aims to achieve and to avoid differences in the application of the model, the choice was to apply the same formula used in some other research (Fair 2007; Ferguson, Jorgensen, and Chen 2019). By using the results of the Republican candidates as the base, the measure is the difference between major party candidates and can be represented as follows:

$$\left[\frac{Rep\ votes}{Rep\ votes + Dem\ votes} \right] - \left[\frac{Dem\ votes}{Rep\ votes + Dem\ votes} \right]$$

The reason that leads to the use of the Republicans as the basis of the formula comes from the fact that it follows the codification of the PVI, that is, the more the value turns out to be positive the more votes the Republican Party has taken, the more negative the more votes the Democratic Party has obtained. In Figure 2 will be possible to observe the election results of the House divided by electoral district, using the scale of the color of the U.S. parties.

⁴⁵ (MIT Election Data and Science Lab 2017, <https://doi.org/10.7910/DVN/IG0UN>)

The results of the 2020 election – Presidential

As known, the results of the 2020 presidential election handed the White House to the Democratic Party and its candidate, former Vice President Joe Biden. The Daily Kow website provides the results of the presidential election, dividing the number of votes obtained by the presidential candidates by the unit of analysis used in this thesis⁴⁶. As was done with the results of the elections to the House of Representatives, also in this case, will be employed the previously mentioned formula which result will be used in the following models. Therefore, through the use of R, there will be a little bit of data processing in order to construct the final variable. Of the four hundred and thirty-six districts that voted in the 2020 elections, that is, all those in the House of Representatives plus the District of Columbia, two hundred and twenty-four voted for Joe Biden while two hundred and twelve voted for Donald Trump with a division similar to that already observed in the results of the lower house of Congress. The results show that the most Democratic district is the District of Columbia where Biden won more than ninety percent of the vote. In contrast, the most Republican district is the fourth in Alabama, where Trump obtained more than eighty percent of the votes cast (Johnson 2021). As was done for the results of the House of Representatives, also in this case there will be a map in Figure 3 to show the distribution of votes obtained by candidates throughout the United States divided by electoral districts. Looking at Figure 2 and Figure 3 at the same time, it is possible to observe how the two are, at least visually, remarkably similar. This could be the first indicator of the absence of ticket-splitting at least during the elections of November 3, 2020. The correlation of colors between the two maps allows to argue that the two results are strongly correlated. So the hypothesis that the presence of a candidate for a more important office helps candidates to win could be valid. For this reason, within the analysis carried out in the following chapter special attention will be given to the results of the presidential elections as they could refute some previous research in this area, such as those done by Born (1994; 2000).

⁴⁶ (Daily Kos Elections et al. 2021, <https://docs.google.com/spreadsheets/d/1XbUXnI9OyfAuhP5P3vWtMuGc5UJlrhXbzZo3AwMuHtk/e dit#gid=0>)

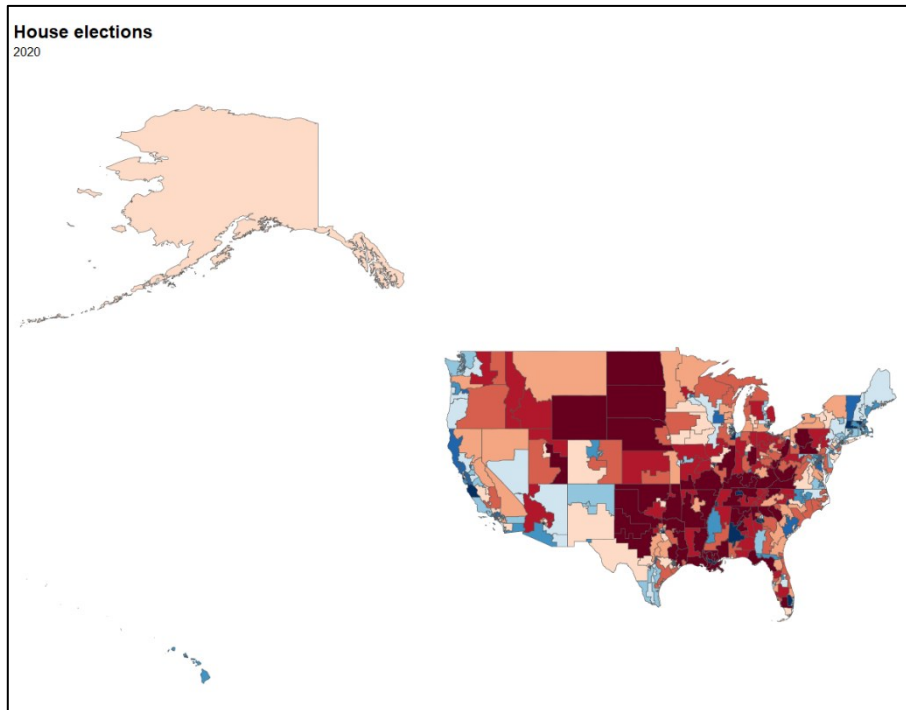


Figure 2 - 2020 House of Representatives election results

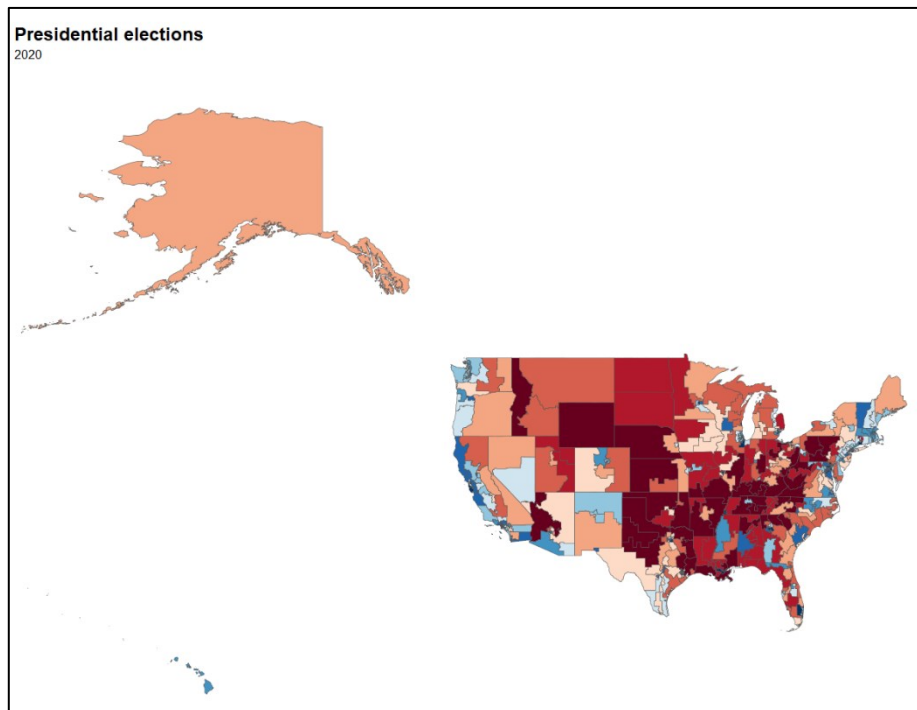


Figure 3 - 2020 Presidential election results

The results of the 2018 election – House of Representatives

The last two elements that must be taken into account within the description of the variable used in this thesis came from the past, because they refer to the election to the House of Representatives of 2018. As for the data regarding the 2020 election, the same data set taken from the Massachusetts Institute of Technology⁴⁷ gave the votes cast in the previous electoral cycle. Thanks to this data set it is possible to create two variables that will be fundamental for the purpose of this work: the outcome of the election in each congressional district and thus the incumbent party in each of them. While the first of the two variables will be used to control reverse causality as set out in the first chapter, the second variable will be a check to see whether or not the election result is influenced by the incumbency status of the candidates, as argued by some authors previously mentioned (Jacobson 2015; Stone et al. 2010). The 2018 elections had given control of the House of Representatives to the Democratic Party which returned to the majority after four legislatures in which they were relegated to the minority⁴⁸. The party division of the 116th Congress numbered two hundred and thirty-five representatives plus the delegate of the District of Columbia registered as members of the Democratic Party and two hundred members within the Conference of the Republican Party. This party division will be functional to consider the incumbency status of the parties in each of the districts. As numerous works cited in the previous chapter highlight, the fact that a candidate is an incumbent or challenger radically changes the way he or she leads the campaign, raises funds, and even spends them. Similarly, the expenses made by independent organizations take this into account because, at the same time, it brings with it some intrinsic advantages such as greater knowledge of the name, a wider network of relationships, and greater experience in Washington. Therefore, the use of a dichotomous variable that indicates whether the Democrats or the Republicans had won two years earlier will try to help understand how much this factor supported the efforts of the candidates in their attempt to elect or re-elect to the Congress in Washington, DC.

⁴⁷ (MIT Election Data and Science Lab 2017, <https://doi.org/10.7910/DVN/IG0UN>)

⁴⁸ (U.S. House of Representatives 2023, <https://history.house.gov/Institution/Party-Divisions/Party-Divisions/>)

Independent expenditures – 2018

As already argued in the previous chapter, the tool that will be used to implement the approach with instrumental variables will be the lagged independent expenditures referring to the electoral cycle in the middle of Donald Trump's presidential term. Also in this case the dataset used will be obtained from the FEC website⁴⁹ and the processing made for independent expenses shown in the first paragraph of this chapter will be re-proposed. Thus, only independent expenditures made to the two candidates supported by the two major U.S. parties will be considered. Also in this case they will be grouped as before by adding the expenses made in support of one candidate and in opposition to the other in order to have a single value for all the candidates present in the elections. As can be argued by looking at the data, the expenses made during this election cycle are much smaller than those made at the same time as the general election. In fact, the average independent spending on behalf of the Democrats is about eight hundred thousand dollars per district, while that of the Republicans is about seven hundred thousand. Also in this case, the distribution on the territory is not homogenous as it ranges from numerous districts in which not a single dollar has been spent, to a district in which the Republicans have spent more than twenty million dollars (GA-06) and one in which the Democrats have spent just under fifteen million (CA-48). As the analysis of Holm, Hjorth-Trolle, and Andersen (2023) argues, the use of a lagged variable as a tool for the approach with IV generally turns out to be the most consistent when it is evaluated through the use of classic measurement error. Since the expenses made by independent organizations are divided between Republicans and Democrats, the lagged independent expenditures taken into account by this paper will also be divided equally. On the contrary, for the expenses made by the candidates' committees, the previous expenses will not be taken into account as these could be linked to other factors such as the quality of the candidates but also the evaluation of the partisanship of the electoral district, as noted by the analysis made by Rekkas (2007).

⁴⁹ (Federal Election Commission 2018, https://www.fec.gov/files/bulk-downloads/2018/independent_expenditure_2018.csv)

	Party	Year	Min	Median	Min	Mean
Independent	D	2020	0.00	3,689.50	12,079,081.00	811,370.50
Independent	R	2020	0.00	6,022.50	12,676,182.00	793,027.90
Candidates	D	2020	0.00	938,397.30	33,728,421.40	1,676,231.50
Candidates	R	2020	0.00	1,079,207.00	22,681,810.00	1,861,640.00
Independent	D	2018	0.00	1,050.50	20,201,362.00	701,787.90
Independent	R	2018	0.00	2,596.50	14,754,552.00	770,661.10
Vote share		2020	-1.00	-0.02	1.00	-0.04
Vote share		2018	-0.93	0.00	0.66	-0.05
Presidential		2018	-0.89	-0.02	0.70	-0.05

Table 1 - Distributions of the variables, all the values have been obtained through the use of R

Method

Through the use of R (Becker 2018) and numerous of its packages, this work will try to understand what the role of independent organizations was and how much they influenced the final result of the 2020 elections. The analysis of the data made in the previous pages is the result of a work of cleaning and arrangement of the data obtained from the various online archives. All of them have been reorganized so that each variable is in correspondence with the district, which means having a data set of four hundred and thirty-six rows, one for each representative, and a cell for each of the fifteen variables used. Starting from what is shown in Table 1 on the previous page, the decision was taken to modify the scale of some variables within the data set. Rescaling functions (Sardy 2008) are essential in data analysis and modeling to bring data into a common scale, ensuring meaningful comparisons and preventing bias. The need for rescaling arises when variables in a dataset have significantly different ranges or units. For instance, variables like vote share and expenditures have different numerical values, which can lead to dominance issues during analysis. For these reasons, the decision was made to scale the funds available to both candidates and independent organizations in millions so that it is better to verify how much the increase or decrease of one million dollars can vary the outcome of the election. This preprocessing step is crucial in this work because it helps avoid undue influence from variables with larger numerical ranges and promotes robust and reliable results. Once the values were rescaled so that the measurement scale was the same for every variable expressing the amount of funds available for the candidates or for the organizations, it was decided to use the three models presented in the introduction of this chapter, an OLS regression adding the extra control variable previously exposed, a spatial regression to evaluate if the distance between districts affects the outcome of the elections, and an IV regression using both the independent expenditures made during the previous cycle as the instrument. Each of these models has the purpose of evaluating at least a part of what is exposed in the first chapter of this thesis. For this reason, in the following pages, the models used will be explained, also listing the statistical tests used to evaluate them.

OLS Regression

Ordinary Least Squares (OLS) regression is a widely used statistical method in econometrics and other fields for analyzing the relationship between a dependent variable and one or more independent variables. It is a simple yet powerful technique that helps us understand the impact of explanatory variables on the target variable. The fundamental idea behind OLS regression is to find the best-fitting line (or hyperplane for multiple variables) through the data points by minimizing the sum of the squared vertical distances between the observed data and the predicted values. In OLS regression, the model assumes a linear relationship between the variables, and it estimates the coefficients of the independent variables to determine the strength and direction of their influence on the dependent variable. This estimation allows researchers to make predictions and draw conclusions about the data. OLS regression serves as a fundamental tool for hypothesis testing, parameter estimation, and assessing the goodness-of-fit of the model, providing valuable insights into the underlying relationships between variables. Moreover, one important aspect will be tested with the use of this regression, which is the possible presence of reciprocal causation that will be tested with the use of the outcome of the 2018 election. A high and significance coefficient in the model will mean the possible presence of an endogeneity problem in the model. After defining all the necessary specifications of the OLS regression and applying the model to the variables previously exposed, it is necessary to perform some statistical tests to evaluate the validity and performance of the model results. The R-squared value will indicate the proportion of variance in the dependent variable explained by the independent variables. An F-test will evaluate the overall significance of the model, determining whether the independent variables collectively have a statistically significant impact on the dependent variable. Additionally, a t-test will be employed to assess the individual significance of each independent variable, indicating whether they significantly influence the dependent variable. Proper interpretation of these statistical values ensures the robustness and reliability of the OLS regression analysis. In addition, it is important to take into account that within the OLS regression the variable concerning the differential of the electoral result of the two parties during the 2018 elections will be inserted to verify the possible presence of reverse causality as previously claimed.

Spatial Regression

Spatial regression is a powerful analytical tool used in various fields, including geography, economics, sociology, and environmental sciences. It addresses the inherent spatial dependence present in many datasets, where observations near each other tend to exhibit similar patterns or relationships. Unlike traditional regression methods that assume the independence of observations, spatial regression accounts for spatial autocorrelation, ensuring more accurate and reliable results. By incorporating spatial factors, such as distance, neighborhood effects, and spatial heterogeneity, spatial regression models enable us to better understand the underlying spatial patterns and interactions in data. These models allow researchers to assess the impact of both traditional explanatory variables and spatially lagged variables, providing insights into the spatial spillover effects and potential spatial heterogeneity. With its ability to capture the complexities of spatial relationships, spatial regression has become an essential tool for decision-making, policy analysis, and spatial planning, contributing significantly to advancing our understanding of spatial processes and phenomena. Starting from all the elements exposed in the first chapter of this work, it seems to be necessary to evaluate the role of spatial distance between districts in order to address the purpose of this thesis. To achieve this result, the first element that must be taken into account in the evaluation of spatial regression is the need to evaluate the presence of some form of spatial autocorrelation between the districts. The fundamental test to address this problem is the Moran's I, which measures spatial autocorrelation in the data. It evaluates whether similar values tend to cluster together or disperse across space. A significant Moran's I test suggests the presence of spatial autocorrelation, indicating that neighboring observations share similar values or almost the fact that it is not possible to reject the presence of spatial autocorrelation in the data used for the research, starting from the results of the OLS regression. The Moran's I test is a spatial statistic used to assess the spatial autocorrelation of data points in a geographical or spatial dataset. It measures whether similar values tend to cluster together (positive autocorrelation) or disperse (negative autocorrelation) across space. It aids in understanding spatial patterns and relationships in data. By using the residuals provided in the OLS model, was run a test using the same matrix that will be used on the spatial regression. Using the *morantest* function on R the results obtained are those

exposed in the following table, in which the main element will be the result of the p-value obtained on the residuals of an OLS that will observe the same factors that the spatial regression should address.

	Value
Standard deviation	-0.42553
P-value	0.6648
Estimate	-5.055481e-03
Mean	-2.298851e-03
Variance	4.196588e-05

Table 2 - Moran I test results

As can be seen in Table 3, the Moran I statistic standard deviation is -0.42553 and the presence of a negative value suggests a tendency toward spatial dispersion or randomness in the residuals. The associated p-value is 0.6648, indicating that there is no strong evidence to reject the null hypothesis of no spatial autocorrelation. This implies that the spatial distribution of residuals may be consistent with a random pattern or a lack of clustering. The sample estimates provide additional context: the Moran I statistic estimate is -5.055481e-03, reflecting the observed value of the Moran I statistic for the residuals. The mean of the tests under the null hypothesis is equal to -2.298851e-03, indicating the value expected if no spatial autocorrelation is present. The variance estimate, 4.196588e-05, quantifies the variability of the statistic under the assumption of no spatial autocorrelation. Based on the results, there is no compelling evidence to suggest significant spatial autocorrelation in the residuals. For this reason, it is not necessary to use this type of regression within the analysis of this thesis and, therefore, the following chapter will only address the two models that use the OLS and the IV regression.

IV Regression

Instrumental Variable (IV) regression is a powerful statistical technique used to estimate causal relationships between variables in the presence of endogeneity, a common issue in econometric analysis. Endogeneity arises when the explanatory variable is correlated with the error term, leading to biased and inconsistent parameter estimates in ordinary least squares (OLS) regression. IV regression addresses this problem by employing one or more instrumental variables, which are exogenous factors that affect the endogenous variable but are unrelated to the error term. The IV approach allows researchers to identify causal effects and overcome potential biases due to omitted variables or reverse causality. It finds application in various fields, including economics, social sciences, and healthcare, where researchers often encounter situations where randomized controlled trials are impractical or unethical. By isolating the true causal impact of a variable of interest, IV regression enhances the reliability of research findings, making it an essential tool for robust and credible statistical analysis. In this case, all the variables applied will be the same as those used for the OLS regression, with the sole exception of the 2018 election result which was to be used to assess the presence of reverse causality in the previous OLS regression. After conducting the IV regression, several key statistical tests will be necessary to assess the validity of the instruments and the robustness of the results of the analysis conducted. For this reason, as was done for the previous model used, even now the analysis will expose the statistical tests used and the values observed to determine the validity of the regression just exposed. The R-squared value quantifies the extent to which the variance in the dependent variable is accounted for by the independent variables. Additionally, by considering all the elements observed through the use of the diagnostic within the summary of the model it will be possible to evaluate the validity of the instruments used for the research. The Weak Instrument Test will help determine whether the two instruments used, namely the lagged independent expenditures in favor of the Democratic party and those in favor of the Republican party referring to the previous electoral cycle, are weak or strong, with weak instruments leading to inconsistent estimates. By performing and interpreting these statistical tests, it will be possible to draw causal inferences with greater confidence when endogeneity is a concern.

Chapter 3 - Analysis

Given that, and the strength of the Cheney name in Wyoming, she has, to my mind, a better than even-odds shot at winning a third term. But money won't win it for her. No matter how much she raises. – Chris Cillizza⁵⁰

This chapter of the thesis will address the three models previously exposed, using the variables mentioned and explained in the previous chapters. The analysis of the models will follow the same structure as the previous chapter: in each paragraph, there will be one of the three regressions built with the use of R, this structure should allow a better comprehension of the work that was done in order to understand the role of the independent expenditures for the election of 2020 to the House of Representatives, but also the influence that the other control variables have on the same outcome. All the models will be based on what has been explained so far in the literature review and in the data and method chapter in which there was a general introduction to the structure of all of them and the statistical values that allow evaluating the results of each of them. As already extensively exposed, the three models used are an OLS regression, an IV regression, and a spatial regression. The results of each regression will be exposed in Table 3 on the following page, with the purpose of better understanding the values that will be exposed through the entire structure of this chapter. Therefore, after carefully observing the values shown in the table on the following page, it is possible to proceed to read the interpretation of the values that will be made in the succeeding pages. After having exposed the results and taking into account what has been expressed so far in the course of this paper, in the last section of this chapter there will be the interpretation of the results based on which have been exposed during the literature review contained in the first chapter of the thesis. In fact, the section named discussion will contain a brief description of what has been found in the outcome of the three models, considering also what has been explained in the previously mentioned literature. Moreover, this part of the chapter will also address the possible limitations of what has been exposed so far, also considering what improvements could be made to the work done.

⁵⁰ (Cillizza 2022, <https://edition.cnn.com/2021/07/13/politics/liz-cheney-fundraising-2022/index.html>)

	<i>Dependent variable:</i>	
	Two-party vote 2020 <i>OLS</i>	<i>instrumental</i> <i>variable</i>
	(1)	(2)
Rep independent expenditures	0.00001 (0.009)	0.100 (0.146)
Dem independent expenditures	0.002 (0.009)	-0.034 (0.088)
Two-party vote 2018	0.100** (0.047)	
Rep candidates' expenditures	0.007*** (0.003)	0.005 (0.005)
Dem candidates' expenditures	-0.012*** (0.004)	-0.043 (0.031)
Partisan Voting Index	0.004*** (0.001)	0.005*** (0.001)
Presidential results	0.768*** (0.061)	0.759*** (0.103)
Constant	0.020** (0.010)	0.026* (0.014)
Observations	436	436
R ²	0.853	0.753
Adjusted R ²	0.850	0.750
Residual Std. Error	0.152 (df = 428)	0.197 (df = 429)
F Statistic	353.921*** (df = 7; 428)	
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table 3 - OLS and IV regression model results

OLS Regression

Using the formula set out in the previous chapter, the dependent variable will then be continuous with negative values if a Democrat won and positive values if a Republican won. Being a differential between the votes obtained by the candidates of the two parties, the smaller the difference between the votes obtained, the lower the value of the two-party vote share. This fact implies that a result with a positive value increases the votes for Republicans, while a negative value increases the percentage obtained by the Democrats. Using all seven variables set out above whose values can be observed in Table 3 on the previous page (independent expenditures in favor of both parties, expenses made by both candidates, the result of the previous elections, the Partisan Voting Index, and the result of the presidential elections) it is possible to observe which are the factors that have most influenced among those examined. The variables encompassed a spectrum of political and contextual factors, ranging from expenditures to historical voting patterns. The obtained results offer valuable insights into the complex dynamics that underpin electoral decision-making. Starting from the two main independent variables, namely the expenses made by independent organizations in favor of each of the candidates, it is possible to observe how both are not very significant, in addition to the fact that in both cases they have very low values because they an increase of one million dollars in both the type of expenditures leads to an increase of 0.00001 for the Republicans and 0.002 for the Democrats, so as to seem almost superfluous for the election of one or the other candidate. On the other hand, one noteworthy finding was the impact of candidate expenditures on the electoral outcome. The regression results revealed that expenditures made by Republican candidates exhibited a statistically significant positive relationship with the two-party vote share of the 2020 House of Representatives election. For every increase of one million dollars in Republican candidates' expenditures, the vote share increased by 0.006. Conversely, expenditures by Democratic candidates were associated with a statistically significant negative effect. A one million dollar increase in Democratic candidates' expenditures led to a decrease of 0.009 in the vote share, increasing the share of the votes obtained by the candidates of this party. These findings underscore the role of campaign finances in shaping voter preferences, suggesting that higher campaign spending may sway voters toward the spending party.

Starting from these two results, therefore, it is possible to see how the expenses made by independent organizations are not so significant with very low values meaning that they will not bring so many new votes, if not given to the opposite party given the fact that the result of those made in favor of the Democrats is positive leading to an increase of the vote share for the Republican party. This type of outcome, which can be seen also in the model using spatial regression, needs to be observed more carefully because, even if the results are not statistically significant, the presence of a positive value for the independent expenditures in support of the Democratic Party needs to be addressed. There are many factors that could have led to such an outcome, of which the three main ones could be the characteristics of the candidates on the ballot, the use of independent expenditures in what was the current political landscape, and the different needs of the U.S. electorate. The first element regards the fact that in 2020 the Republican party reached a record number of women and members of ethnic minorities as candidates over the whole country (Weissert and Beaumont 2020) changing the local perception of the two parties that compose the political system. The presence of this type of candidate could have created a sort of distortion in voters' reception of messages transmitted by independent organizations mainly because these two categories usually tend to vote more for the Democratic party. The use of general messages that do not refer to any specific candidates but just to women and ethnic minorities could have driven citizens not to vote for the party that they want, but for the candidates that can be defined by these characteristics. The second factor refers to an element that was previously mentioned in the first chapter of this thesis. Since the Roberts Court struck down the McCain-Feingold Act in 2010, the use of independent expenditures favored better election results for the Republican Party. Due to this fact, it can be possible to explain why also those made in support of Democratic candidates have led to an increase in the vote shares obtained by the Republicans. The last element that needs to be considered is slightly more complicated because it refers to the public perception of the two parties. By observing the results of the exit poll conducted right after the end of the elections⁵¹, it is possible to notice that there is a crucial question that is *“Most important issue in your vote for president”* that is divided into five

⁵¹ (Presidential exit poll, <https://www.washingtonpost.com/elections/interactive/2020/exit-polls/presidential-election-exit-polls/>)

questions, namely Covid, economy, crime and safety, healthcare, and racial inequality. Crossing these data with what emerged from the Pew Research Center research⁵², it is possible to see how the majority of citizens consider the economy as the main priority of the government and, for this task, they rely more on the Republican Party. Using economy-focused advertising, therefore, may have led the electorate to the party they considered most competent to fix this problem and not to consider the message conveyed in support of which party. On the contrary, the expense made by candidates would seem to play a fundamental role in the electoral process of candidates for the House of Representatives. Moreover, the analysis also delved into the impact of a broader political context. The Partisan Voting Index, which was coded as the two-party vote share using positive values for the Republicans and negative for the Democrats, was found to be a significant determinant, indicating that a positive partisan-leaning district tends to increase the votes received by the GOP (0.004). Similarly, also the presence of the presidential election seems to have an extraordinarily strong effect on the electoral results, suggesting that the constant presence of the ticket-splitting phenomenon throughout the last century is now overcome by factors such as political polarization. The results of the Presidential election exerted a profound influence on the two-party vote share in 2020. A one-unit increase in the two-party vote share for the presidential results leads to a remarkable 0.725 increase in the vote share in the share of the House of Representatives. These particularly important findings suggest that national political sentiment is driven by the presence of both Trump and Biden at the top of the ticket. Moreover, the polarization of the political landscape has led to a sharp division between the electorate of the two parties that would now seem to prefer a solid majority in the elective branches of a single party, rather than a balance of power as was the case until the last decade of the twentieth century. The last element taken into consideration within the model is the two-party vote share referred to 2018, which is the electoral cycle in the middle of Trump's presidential term. The historical voting patterns exhibited a substantial influence on the share of the votes obtained by both candidates in 2020. The vote share in the previous election displayed a statistically significant positive relationship with the 2020 vote share (0.100). This

⁵² (Pew Research Center 2021, <https://www.pewresearch.org/politics/2021/01/28/economy-and-covid-19-top-the-publics-policy-agenda-for-2021/>)

implies a degree of inertia in voter behavior, where past electoral outcomes contribute to shaping current decisions, meaning that the previous election result could be one of the factors driving the vote in the next cycle. Among other things, this factor implies that there could be a problem of endogeneity caused by the reciprocal causation between the outcome of the election, which is the dependent variable, and the two main independents, the independent expenditure made in favor or against the candidates, since the expectation of victory could increase the funds received by the candidates. The strong validity and value of the result of the previous elections lead precisely to imagine that these two factors can be linked, justifying the use of an IV regression among the models used by this work. Looking at the values of the significance of this regression, it is possible to analyze the value of the R-squared and of the F-statistic. These two statistical elements allow to evaluate the results provided by the whole model, giving the possibility of saying if the model is significant or, otherwise, it needs to be improved. Therefore, thanks to those two factors, it is possible to address the quality and the reliability of the whole model, but also of the single observed elements. The R-squared value of 0.859 signifies that approximately 85.9% of the variability in the two-party vote share in the 2020 election can be explained by the variation in the chosen seven independent variables previously observed. This high R-squared indicates a robust fit of the model to the data. The F-Statistic of 323.836, with degrees of freedom of 8 and 427, highlights the overall statistical significance of the model. The extremely low p-value associated with this statistic indicates that the collective impact of the independent variables on the differential between the number of votes received by the Republican party and the Democratic party in 2020 is highly significant. These supplementary statistics fortify the analysis, providing a comprehensive assessment of the model's fit, accuracy, and significance. They collectively affirm the model's reliability in capturing and explaining how it was possible to obtain that particular result measured with the two-party vote share. Thanks to these values it is possible to affirm that the results of this model are good enough to confirm the results previously expressed. Although the value of the expenses made by independent organizations is low enough to reject the fact that they played a crucial role during the elections, it can be said that the expenses made by the candidates contributed to the final result, increasing the votes for both parties in the analysis.

IV Regression

Thanks to the fact that, as seen in the previous model, there is the possibility that there is a reciprocal causation between the dependent variable and the independent one, in this case, the model with the IV regression will be used to try to overcome the risk of endogeneity present. As explained above, the instruments used will be independent spending for both candidates during the previous election cycle. As has been amply explained throughout the first chapter, the expenses made by independent organizations during the 2018 electoral cycle cannot play a role in the 2020 result both because they are not made by the same organizations, and because the funds used are different, but also because there is a large turnover of members of the House of Representatives between the elections. The IV regression model offers an innovative approach to tackle endogeneity concerns that may plague conventional linear regression. By utilizing instrumental variables that are correlated with the independent variables but not directly with the dependent variable, the aim is to disentangle causal relationships that might otherwise remain elusive. As in the previous model, also in this case the dependent variable will be the two-party vote share of the 2020 election with the same assumptions set out above, while the list of independent variables will include all the previously applied variables, with the lonely exceptions of the outcome of the 2018 elections. As was done for the previous model, the first two elements that will be observed are the two main independent variables, the independent expenditures made in favor of both parties. Campaign expenditures have long been recognized as a key driver of electoral success. In the model exposed in Table 3, Republican and Democratic independent expenditures are evaluated for their influence on the two-party vote share of the 2020 elections. The results reveal intriguing patterns. Republican independent expenditures exhibit a positive coefficient of 0.100, albeit not statistically significant. This suggests that, while higher Republican expenditures are associated with increased vote shares, the observed effect lacks statistical robustness. On the Democratic side, and differently from what was found out in the OLS regression, the coefficient for independent expenditures is negative (-0.034), though also statistically insignificant. Again, these results raise some doubts about the importance of independent expenditures made during the 2020 cycle for the House of Representatives elections. Moving beyond independent expenditures, it is necessary

to scrutinize the contributions of candidate-specific spending. Notably, Republican candidates' expenditures have a positive coefficient of 0.005, implying that higher spending by Republican candidates corresponds to a higher value of the two-party vote share. Similarly, Democratic candidates' expenditures exhibit a negative coefficient of -0.043, indicating that higher spending by Democratic candidates is associated with a lower value of the same differential, meaning an increase in the number of votes received by this party. Therefore, differently from what was observed in the first model, none of the expenditures seems to have a crucial and leading role in the outcome of the elections. On the other hand, and similarly to what was observed in the OLS regression the contextual factors, including the Partisan Voting Index and presidential election results, are central components of the analysis made. The PVI coefficient is positive (0.005) and statistically significant, suggesting that the more a district leans toward the Republican Party, the more votes the GOP candidates will receive, and, at the same time, the more the PVI is lower and tends towards the Democratic Party, the more votes the candidate of this party will receive. This underscores the idea that local political leanings resonate with voters and impact their choices. The variable containing the two-party vote share for the presidential results yields a substantial coefficient of 0.759, demonstrating a powerful influence. A one-unit increase in presidential results, which means an increase in the votes received by Trump, corresponds to a substantial increase in the dependent variable of the analysis. At the same time, a decrease of one unit in this variable, which means an increase of one share point for Biden, leads to an increase of the share obtained by the Democratic candidates to the lower chamber of Congress. This finding highlights the resonance of presidential election outcomes and their ability to sway subsequent elections. Notably, the coefficient carries a high degree of statistical significance, reinforcing the robustness of the observed relationship. The results of this model show the centrality of the role played by the presidential elections and the historical-political trend of the electoral district. Also in this case, it is necessary to go to observe the statistical values expressed at the end of the same table that contains the results of the model to try to understand if the model is actually efficient. The regression model's overall fit is assessed through the R-squared, which stands at 0.753. This statistic indicates that the model explains approximately 75.3% of the variability in the two-party vote share of

the 2020 elections. While a substantial proportion of the variance is captured, it's important to recognize that other unmeasured factors contribute to electoral outcomes. The residual standard error of 0.197 provides insight into the average distance between observed and predicted values. A lower residual standard error indicates a better fit of the model to the data. Moreover, Using the results of diagnostics in the observation of the result of regression IV, it can be observed that in both p-values of both instruments, meaning that both the independent expenditures made to support the candidates of the Republican Party and those made to back the Democrat candidates in the 2018 election cycle are quite good instruments for the purpose that the model wants to achieve, also if the expenditures made to promote the candidates of the second party have the lower p-value between the two instruments. Thanks to the fact that all the values presented here, both the result of the R-squared, the residual standard error, and the values of the two tools, it is possible to argue that the results of this model are statistically valid and, therefore, it is possible to affirm again that also, in this case, the model is statistically valid. Moreover, also in this case as what was done within the analysis of the first regression, it is necessary to underline the contractual role played by the presence of the elections to the Presidency of the United States that would seem to have driven the competition for seats in the House of Representatives throughout the country. At the same time, the party bias of each district would also seem to have played a crucial role in defining the composition of the lower chamber of Congress after the autumn 2020 elections. According to the model with regression IV, these two factors are the most significant and statistically valid for the analysis that this thesis aims to do. On the contrary, any kind of expenditure, whether made by the committee led by the candidates or by the numerous independent organizations, would seem to be irrelevant for the purposes of assessing the increase in the votes of both the Republican and the Democratic parties. Moreover and unlike what was found out in the previous model, it is possible to note that in this case, all the expenses made have the corresponding mathematical signs equal, meaning that both the expenditures made by the Republicans have a positive effect on the two-party vote share, while both the Democrats have a negative effect. This element is noteworthy, especially because, as can be seen from the previous Table 3, this model is the only one of the three in which this event occurs. Moreover, in the case of the Instrumental Variable regression, it is

necessary to evaluate three more elements inserted within the diagnostic test section, which includes the results of various diagnostic tests to assess the validity of the instrumental variables and the overall model fit. The three available tests are the weak instrument, the Wu-Hausman test, and the Sargan test. The analysis indicates potential issues with weak instruments in the model, as suggested by the diagnostic tests. The p-values associated with the test for the research of the presence of weak instruments, which consider both the lagged independent expenditures of the 2018 midterm electoral cycle provided to support both the Republican and the Democratic candidates are 0.059 and 0.000196, respectively. These values suggest that in the first of the two cases, the instrument might not be sufficiently strong to address endogeneity concerns while taking into account the lagged expenditures provided to support the Democratic Party seems to be quite efficient, allowing to avoid the presence of endogeneity. This raises caution about the reliability of the instrumented regression results and suggests the need for further evaluation or potentially more suitable instruments for a robust analysis. The Wu-Hausman test outcome is sufficiently valid because of the results of the p-value which is equal to 0.438. This test examines endogeneity, which refers to the potential correlation between independent variables and the error term. The test result implies that the presence of endogeneity is not statistically significant in the model. In fact, incorporating lagged independent expenditures as instruments in IV regression enhances causal inference by addressing endogeneity. Time-lagged variables mitigate reverse causality, aiding in the identification of a robust relationship between the endogenous variable and the instrumental variable, thereby bolstering the validity of the results. The last statistical value examined is the one provided by the Sargan test, which in the analysis gives just an NA. This test is used to check the over-identifying restrictions in the model. Due to the absence of a p-value of this element, It is possible to exclude the slightest influence of this factor within the analysis presented so far. However, it's important to interpret all those values taking into account that it is necessary to evaluate all the other elements within the analysis previously exposed. All the findings exposed so far have almost tried to underscore the intricate interplay of financial efforts, provided by independent organizations and candidates for the U.S. House, historical patterns, and contextual factors in shaping voter behavior and electoral outcomes during the 2020 elections.

Discussion

Taking into account what has been explained so far, the role of independent expenditure would seem to have been quite superfluous given the statistical non-significance of the results obtained in all three models. As has been explained, the reasons why independent spending in support of Democratic candidates has a positive value causing a hypothetical, even if not very significant, damage are basically three. They are the characteristics of the candidates on the ballot, the tendency in the use of independent expenditures in what was the current political landscape, and the different needs of the U.S. electorate. The first factor considers the presence of a historical record number of women and ethnic minority candidates for the Republican Party, while those two groups tend to vote more for the Democratic Party. The second element depends on the fact that wealthy individuals tend to vote more for the GOP, and the large majority of them will have the possibility of contributing a large amount of money to the organizations that support this party. The last element considers the general belief that the Republicans are more able to address and fix the economic problem that the United States was facing and, therefore, the use of advertising that talks about the economic situation will increase the number of votes for the conservatives. These factors, noted by some analysts (Magleby 2014; Ward 2016), may have supported, even if minimally, the fact that the expenses made by independent organizations in favor of the Democratic candidates, in fact, ended up favoring those of the opposite camp, contributing to the reduction of seats for the party led by Nancy Pelosi. In reality, however, the influence of these expenses seems to be very low, and, therefore, even their role would seem to have been only marginal. On the contrary, the role played by the expenses made by the candidates' committees themselves would seem to be central to the analysis of this thesis given the high significance of the results expressed by the p-value and the much more relevant value that these have, as shown in Table 3. Except in the case of regression IV, in fact, the expenses made by the candidates have a much higher and even significant value, and that allows to affirm that these expenses played a central role during the 2020 elections for the Chamber of Representatives. This fact confirms the findings of research by Giertz and Sullivan (1977), who argue that the spending made by candidates is able to influence the behavior of the electorate during the elections. Moreover, the expenses of both

Democratic and Republican candidates have a sign – negative and positive respectively – which indicates precisely the fact that they increase the votes of their respective parties during the general elections. The contribution made by this type of expenditure is crucial in understanding the 2020 election result. So, thanks to what has been achieved at least in the model with the OLS regression, the increase of one million dollars of expenditure by the candidates goes in the direction of helping them get elected. Therefore, this fact confirms the findings of previous research (Potters, Sloof, and van Winden 1997) about the role of the expenditures made by candidates within the U.S. electoral system. The other elements observed look at the historical and political factors that may have played a role in shaping the outcome of the election. The Partisan Voting Index of the districts turns out to be fundamental in each of the models exposed, confirming that the more a district tends to vote for a party, the more this party tends to gain votes. The use of PVI as a continuous variable with negative values for Democrats and positive values for Republicans as was the distribution of the two-party vote share allowed to understand when the polarization of the districts helped the candidates to get elected. An important factor in this case is the fact that it is often the local legislators themselves, those of the houses and senates of each state, who draw the districts for elections to the House of Representatives taking into account the historical results in each county so as to be able to favor their party during the elections by applying the Gerrymandering system, as noted by the analysis of Wang, Ober, and Williams (2019). Therefore, the use of PVI also allows to consider this factor that is central to the U.S. political system. Considering the result of the two previous presidential elections, the PVI allows both to evaluate the polarization of the districts and also to consider how much historical factors supported the victory of those who would later become the members of the House of Representatives. The other historical-political factor taken into consideration is the role played by the simultaneous presence of the presidential elections. In all three models, the p-value for the variable concerning voting in the presidential election is highly significant and its value is extremely high. As was expressed previously, the use of this control variable was due to the fact that during the whole XX century, there was a general tendency to vote for the Republican as President, while having a Democratic majority in Congress. After the first midterm election during the presidency of Clinton, the Republicans gain the majority in

Congress and since that year the presence of ticket-splitting decreased. All the elections since the beginning of the new century were characterized by this factor, even if there were some more peculiar outcomes like the elections of 2012 when Obama won reelection, but Republicans kept control of the House, the election of 2016 when Trump won the presidency, Republicans kept control of the House but lost some seats. As can be seen in Figure 4, there is a strong relationship between these variables because when the votes share for the Republican presidential candidate increases, also the share of the candidates for the House of Representatives will increase.

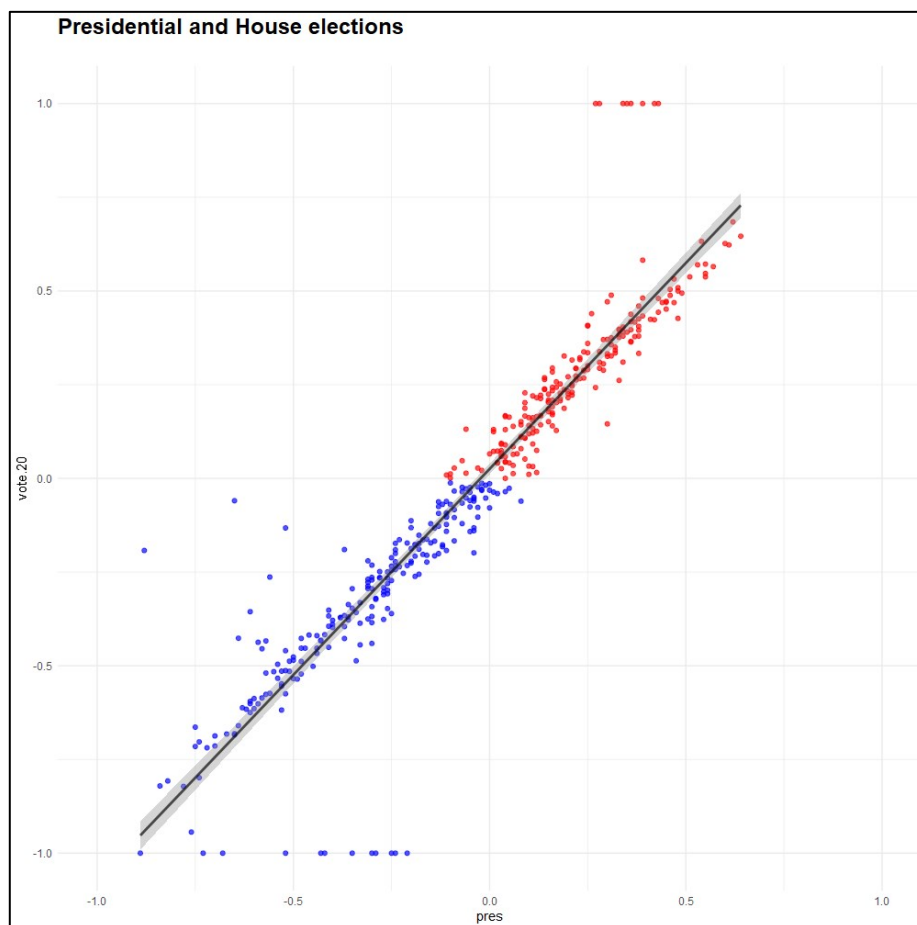


Figure 4 - Presidential and Congressional results

Thanks to the results expressed in Table 4 regarding the relationship between these two variables, it is possible to confirm what has been stated by Burden and Kimball (1998) about the fact that the outcome of the Presidential election could be the element that has the large influence on determining the outcome of the elections to the U.S. Congress. As expressed by Kuriwaki (2023) *“the nationalization of voter*

behavior in recent decades is thought to have shifted the electoral landscape” (p. 3) which is the central element that causes the use of such a variable in this thesis. The nationalization of electoral campaigns and advertising used by congressional candidates, combined with a growing polarization of the U.S. political system, could have determined the outcome of the 2020 elections to the House of Representatives. The last element that has to be addressed is the outcome of the 2018 midterm elections. The use of this variable was due to the fact that there was the possibility of having reverse causality between the dependent and the two main independent variables, namely the expenditures made by independent organizations. As can be seen in the column regarding the OLS regression in Table 4, the previous electoral result has a significant impact on the outcome of the following. The significance of this variable is determined by a high p-value and, therefore, it is possible to express some form of concern regarding the presence of reverse causality, justifying the use of the IV regression in the second model of this thesis.

In conclusion, it can be said that, given the results of the three models set out above, the role of spending by independent organizations has proved to be insignificant if not harmful in the case of democrats, while, on the contrary, the role of candidates' expenses is central within the electoral process. Even more central is the role played by the party tendency of each district and the simultaneous presence of elections for national offices, as precisely that of the President. Therefore, the role played by these factors is decisive in understanding how an electoral result of this type was arrived at with the 2020 elections. In any case, it is vital to acknowledge that all three models represent a simplified abstraction of the complex electoral landscape, and further research can refine the understanding of the factors that drive democratic participation and choice, allowing to better understand the role played by each of these, but it would also be necessary to add other factors that may have played a role in arriving at this type of outcome. In fact, taking into consideration what has been explained so far, it is possible to observe the presence of some limitations in the work done. The first limitation that must be highlighted is the lack of descriptive factors of the local electorate. Elements such as ethnic and gender composition, wealth, level of education, but also all other hypothetical descriptive variables of districts could help in understanding the distribution of votes within each electoral college. All of these

factors played a crucial role in determining the outcome of the election, especially as polarization has led to certain constituencies tending to vote with a clear majority for only one of the two camps. As can be seen by observing the exit polls of the elections under consideration⁵³, these are the first characteristics that are illustrated by analysts, but they are also the basis of the political strategies of the parties, as they allow to better define how a certain candidate, or alignment, can increase his chances of being able to win the electoral competition. Therefore, defining these characteristics would permit to understand why a particular candidate can win in a district especially if only with a very narrow margin. But not only do the demographic characteristics of the districts help to define the election result, but other elements at the local level can also allow a better understanding of the outcome of the elections. A clear example would be the approval rating of both parties and all types of candidates. A high level of approval from the incumbent President or Representative could help the party in the local competition, as it would decrease the chance that voters will decide to switch sides from the previous election cycle, rewarding the work they have done over the past two years. Considering these factors could make it easier to understand this type of electoral outcome. but also the use of descriptive variables of the candidates, both from the point of view of ethnicity, gender, and personal wealth and from that of previous work experience. Considering a candidate who has previously faced different electoral contests in the same way as one who is in his first candidacy could avoid checking for an important factor. In addition, a further limitation could be to consider all electoral districts, regardless of their own characteristics. Equating a district in which the incumbent candidate is running for re-election with one in which the representative has decided to withdraw could create some problems. In fact, as has been repeatedly said, the electoral strategies of the candidates are different even, if not especially if they are in office, challengers, or if they run in an open district. For the purposes of this thesis, it was necessary to evaluate all districts, but further research should take this factor into account. For this fact, it would be useful to introduce a tool for the evaluation and classification of districts, if not even a selection of which districts to consider and which, instead, to remove from the analysis that must be

⁵³ (Presidential exit poll, <https://www.washingtonpost.com/elections/interactive/2020/exit-polls/presidential-election-exit-polls/>)

carried out. A further limitation of this work concerns the result it was intended to achieve. Taking into account the fact that the expenses made by independent organizations would seem to be irrelevant to the election results, it must be taken into account that in many districts considered uncompetitive these organizations have not made any type of expenditure, involving the fact that there is a large number of constituencies reporting zero in the two main independent variables. This could lead to a significant reduction in the role of independent expenditure in the electoral system and it would be necessary to take action in this regard to better assess their role throughout the process. For this further reason, a classification or reduction in the number of districts may be necessary to increase the value of the work done. A further element could concern the timing of data collection. The work carried out during this thesis took only the final total of the expenses made without considering the historical progression, therefore without evaluating when these expenses were made from the day after the primary election to the day of the November election. Consider whether during this time there has been a moment of increase or decrease, combined with the presence of external factors such as scandals, could help to understand the final result. The last element that should be taken into account concerns the left-right orientation of the candidates. A candidate with more moderate positions, in fact, will have a better chance of being elected by being able to collect votes even in the opposite camp. A clear example of this is the re-election of Senator Collins in 2020 despite all the polls showing her largely losing (Berman 2020). At the same time, the orientation of donations, as pointed out by Bonica (2016), could also play a role and would be an element to be taken into account in any future research.

Conclusion

*Republicans actually gained seats in the House, by the way. So cue the Democratic finger-pointing. We've already heard and seen some. Progressives and moderates blaming each other for the results. – Chuck Todd*⁵⁴

The comprehensive analysis conducted in this thesis has yielded a profound understanding of the pivotal role assumed by expenditures originating from independent organizations, most notably the Super PACs and the 501(c)s, in shaping the outcomes of the 2020 House of Representatives elections. The two main independent variables adopted by this thesis, independent expenditures in support of Republicans and Democrats respectively, have undergone meticulous scrutiny within the models constructed in the preceding chapter. The goal set by this thesis was precisely to understand how much the millions, if not billions, of dollars spent by independent organizations have managed to move in terms of percentage points on the two-party vote share, the differential in votes obtained by the two main parties of the U.S. political landscape. The thesis' regression analysis results offer some very interesting hints. Starting from the literature widely presented and reviewed in the first chapter of this work, the theoretical basis from which the analysis carried out by this research starts has been defined, both in terms of variables used to carry out the analysis and in terms of the models used to evaluate those same elements. By employing the two-party vote share as a dependent variable and the expenses made by the independent organizations supporting the two parties as the main independent variables, an attempt was made to understand their role, while evaluating the role played by the expenses made by the candidates' committees, the votes for the President of the United States, the partisan division of the districts, as well as the results of the previous elections for the House of Representatives, namely those of 2018. Therefore, starting from the results presented in the previous chapter, it is possible to draw some conclusions about the role played by this type of expenditure, as well as that of the other variables considered and mentioned in the previous lines, in determining the outcome of the elections for the House of Representatives in 2020.

Starting from what has been shown in the previous pages, it is possible to identify some interesting elements in the results regarding the expenses made by both

⁵⁴ (Todd and Clyburn 2020, <https://www.nbcnews.com/meet-the-press/meet-press-11-08-20-n1247045>)

parties. For Republican candidates, the variable concerning the independent expenditures indicates a positive relationship with the two-party vote share. However, the results exposed in Table 3 suggest that the impact of independent expenditures on the Republican vote share is minimal. In contrast, the independent expenditures made in support of Democratic candidates show both a positive and a negative relationship with the two-party vote share. In this case, unlike what was obtained with the expenses in support of the Republicans, the results would seem to imply that independent expenditures might have a slightly stronger impact on the Democratic vote share. The element that is of greatest interest is the fact that the results for the independent expenditures are all positive, both those for expenses in support of Republican candidates and those in support of Democratic candidates. This suggests that, within the range of the data analyzed, the influence of independent expenditures on the two-party vote share is limited. Since the two-party vote share was coded using the votes obtained by the Republican candidates as a basis, obtaining a positive coefficient in the models means that when the organizations supporting the Republicans spend a million dollars more there should be a growth in the differential considered, while if those in favor of the Democrats do the same value should, at least in theory, go down. As the OLS regression shows, the spending values made by independent organizations in support of the Democrats would seem to favor candidates on the opposite side. The explanation for this phenomenon, as mentioned in the previous chapter, lies in the presence of some important factors such as the candidacies of women and members of ethnic minorities, the fact that independent spending tends to be more favorable to Republicans as claimed in the literature, and greater confidence of the electorate in the way the GOP handles economic problems than what the Democrats did. These, combined with other factors that may not have been considered, such as the approval rating of the candidates, their positioning on the political axis, or their political and life experience, may have played a role in determining the impact of spending on the November 2020 House of Representatives elections.

The second element that must be taken into account is the role played by the expenses made directly by the electoral committees managed by the candidates for seats in the House of Representatives. As aforementioned, there can be no coordination between candidates and independent organizations and, therefore, it is also necessary

to control the role played by this type of expenditure within the electoral process. Employing an OLS regression model, this analysis found that an increase in the Republican candidates' expenditures corresponds to an increase in the two-party vote share, whereas an increase in the Democratic candidates' expenditures corresponds to a decrease in the two-party vote share. Furthermore, the findings are corroborated by the IV regression, whose results are consistent with the OLS regression. The results are consistent in both models employed, suggesting that the association found between the dependent variable, the two-party vote share, and the independent one, the expenses made by independent organizations, actually occur regardless of measurement approaches. These results collectively indicate that campaign expenditures do indeed influence the two-party vote share in the direction foreseen by the structure of the dependent variable. Therefore, given the values obtained in the models and the analysis presented so far, it is possible to argue that the expenses made directly by the candidates' committees were not only much higher but also, at least in the case of the OLS regression, highly significant. The fact that these expenditures were more important than those made by independent organizations can be interpreted in different ways. Among them, the fact that it is the candidates who manage them directly, the fact that the expenses made on advertising go directly in the direction of supporting a candidate, or the fact that independent advertisements are really only a mimicking of messages already broadcast by official election campaigns.

The last elements that have been taken into account are those referring to the surrounding political environment and resulting from the previous elections for the House of Representatives (which is used only in the OLS regression), the presidential election, which is always calculated as a two-party vote share with the Republicans as a base, and the Partisan Vote Index (PVI) of the districts, which refers to the results of the two previous presidential elections in each constituency. The result of the previous electoral challenge is very significant, indicating that the vote cast during the previous election cycle influences the vote that will be cast during a future election, rather than both types of expenditure already exposed. This result also justified the application of model IV, which was based on the possibility of observing reverse causality between the dependent variable and the two main independents of this analysis. Taking this value into account, victory in the previous electoral cycle could be one of the elements

that most influence the possibility of victory in the next electoral cycle, supporting the fact expressed by numerous previous researches that incumbent candidates start as favorites in the challenge for re-election. The second factor considered, the PVI, indicates the frequency with which the district in question tends to vote for one party or the other. Also in this case the variable has been coded with positive values in the case of votes more for the Republican Party or negative values if it is more aligned with the Democratic Party. In both models, this value is highly significant, and that means that the more partisan the district tends to be, the more it will tend to vote for that party's candidate. The last element considered is the one that turns out to be the most influential among the variables used by the analysis. Looking at the values shown within the same table, it is possible to see how the result of the presidential elections codified as two-party vote share always with the Republicans as a base is the variable that most explain the outcome of the elections for the House of Representatives. The element that turns out to be the most central within the analysis carried out by this thesis is the vote obtained by the presidential candidates in each of the electoral districts. The results shown for these variables in Table 3 mean that voting for Biden or Trump is a good indicator of understanding the vote for Congress. In fact, the fact that the presidential election ended with a fairly large margin of votes does not indicate that the two candidates came in almost equal in the number of districts won as shown in Figure 4.

In conclusion, once the results presented so far have been carefully observed, it is possible to say that independent spending would not seem to have played a crucial role in determining the outcome of the elections for the House of Representatives in 2020. On the contrary, all the other variables have higher values and are almost always significant, showing how these could have had a greater incidence than the two main elements considered by this thesis. Therefore, there would not seem to be a significant role on the part of the expenses made by independent organizations, although these are still very high. The research conducted so far has tried to define the role of independent organizations in a single electoral cycle by trying to use the tools of literature and considering, unlike what has been done by most of the works analyzed, this type of expenditure. Given the results obtained, however, these expenses would seem to have been inefficient and, therefore, it could be interesting to expand knowledge in this area

by observing other elements that may have played a role in determining the actual value of this type of expenditure, such as considering only expenditures made in the most competitive districts. The extensive array of innovations and constraints that have been highlighted throughout the course of this thesis underscores the immense complexity and challenge inherent in comprehensively assessing this subject matter. This assertion reflects the multifaceted nature of the topic, which defies simplistic categorization or evaluation. Throughout this research, there were proposed a diverse range of innovative ideas, methodologies, and perspectives that have significantly enriched our understanding of the subject at hand. These innovations span various disciplines, from technology and science to social sciences and humanities, and have provided valuable insights into the intricacies of the topic. On the other hand, it is equally important to acknowledge the inherent limitations that have arisen during the course of this inquiry. These limitations, whether in terms of data availability, methodological constraints, or theoretical frameworks, have posed challenges to our efforts to comprehensively evaluate the subject matter. They serve as a reminder that no research endeavor can fully encapsulate the entirety of such a vast and intricate topic. The thesis has illuminated the nuanced nature of the role of independent expenditures, offering a glimpse into its richness and complexity. While the innovations have expanded the knowledge, the limitations underscore the need for continued exploration and refinement in understanding.

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